

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics

In partial fulfillment for the requirements in Capstone 1 and
Research Subject

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CHAPTER 1

1.1 Background of the Study

In today's fast-paced world, businesses across all sectors are shifting to online platforms to keep up with the growing expectations of their consumers. The food service and enterprise baking industry is no exception to this trend. As tech-savvy individuals increasingly turn to digital applications for seamless, on-demand transactions, local micro-enterprises that continue to rely on manual, analog processes face significant challenges. Traditional, non-automated methods often result in critical operational inefficiencies, reduced market reach, and an inability to meet contemporary consumer demands. This shift in buyer behavior has turned digital convenience from a luxury into an absolute necessity for survival in the food retail industry. When small storefronts lack a modern web presence, they hiddenly lose customers to larger competitors who offer instant online ordering.

The underlying problem stems from how consumer data is collected and managed during initial touchpoints. In a typical digital commerce environment, data inputs are structured, validated, and processed instantaneously by backend application logic. Conversely, traditional storefront structures rely heavily on human memory and fragmented paperwork to handle transactions. This lack of automated validation means that business transactions are highly susceptible to data loss, tracking delays, and operational blind spots. For community-scale enterprises, continuing to operate within a purely analog framework creates a technical barrier that cuts them off from

the modern digital marketplace, making it increasingly difficult to sustain consistent growth or build long-term customer loyalty.

C&N Bakery is a local bakery operating within a highly competitive local market, offering a wide selection of baked goods categorized into standard product groups, namely cakes, cupcakes, and traditional Filipino kakanin. In addition to mass retail pastries, the establishment specializes in bespoke, high-value custom cake orders and supports a pre-ordering framework designed for clients requiring specialized baked goods for celebrations, weddings, debuts, and milestones. Despite managing a diverse product line and catering to a growing clientele, C&N Bakery handles its entire business workflow through manual mechanisms, such as physical walk-ins, phone calls, and scattered social media messages, without any centralized digital infrastructure to sustain its operations. This lack of a single system means information is scattered across different mediums, making daily tasks stressful and disorganized.

When an information system lacks a centralized architectural hub, data fragmentation becomes a persistent operational issue. At C&N Bakery, a single custom transaction often leaves a scattered data trail across multiple platforms: a Facebook Messenger text thread may contain the cake's design layout, a follow-up phone call might alter the delivery deadline, and a hand-written paper note on the counter might record the downpayment. Because these data points are completely unlinked, the administrative staff must manually piece together the customer's files before sending the order to the kitchen. This fragmented approach consumes massive amounts of time and makes

day-to-day operations chaotic, turning simple routine tasks into exhausting manual procedures that drain the small business's daily energy.

The administrative overhead of managing custom cake orders is inherently more complex than standard point-of-sale retail transactions. Customizations demand precise recording of multiple structural variables, including layers, chiffon flavors, frosting variations, visual design styles, and custom dedication text, alongside strict coordination of pickup and delivery dates to preserve food freshness and structural integrity. Relying on handwritten logbooks, paper receipts, and unorganized chat logs severely increases the risk of order leakage, a critical vulnerability where explicit structural design choices are forgotten or overlooked during preparation. Furthermore, human error in manual calendar logging leads to schedule overlaps, where production capacity is inadvertently exceeded for a specific date.

This operational vulnerability is deeply tied to the technical concept of data mutation. When a customer outlines their custom cake selections through a social media chat channel, that information exists as an unformatted text string. As that request moves from the front counter staff to the kitchen bakers, the information is manually rewritten onto paper order tags. Every time a human manually copies, interprets, or rewrites a data string, the higher the risk of transcription mistakes occurring. Essential design attributes—such as specific icing colors or custom text names—can easily be misread or completely forgotten during the transfer. This manual workflow creates a weak system loop where the original intent of the customer rarely matches the actual physical instructions given

to the baking staff.

When the bakery relies on scattered chats and unorganized paper notes, it creates a high risk of transcription mistakes during the order-taking process. These data entry errors directly lead to incorrect cake designs being sent to the kitchen. Once a cake is baked with the wrong design, flavor, or configuration, it triggers a chain reaction of operational failures across the business. This workflow breakdown results in wasted raw ingredients, disrupted baking queues, and unrecoverable financial losses for the bakery, while severely damaging long-term customer trust.

The financial damage caused by these data errors is made worse by the rigid nature of the baking process itself. Unlike other food service businesses that can quickly cook a replacement dish in a few minutes, a custom cake requires hours of uninterrupted baking, cooling, structural stabilization, and detailed frosting application. If a kitchen assistant discovers a configuration error late in the pipeline due to a manual data entry mistake, the entire product must often be discarded. This means the bakery loses both the expensive ingredients used and the valuable time slots in the oven schedule. This disruption delays every other order waiting in the queue, creating a stressful work environment and causing a backlog of delayed deliveries that ripples through the entire business day.

The operational difficulties are further worsened by the rapidly evolving commercial ecosystem in the Philippines, marked by the widespread adoption of e-wallets and digital banking applications. Modern consumers expect transaction flows that offer instantaneous price validation, absolute cost transparency, interactive customer review modules, and

structured frequently asked questions (FAQ) components, combined with the convenience of settling accounts immediately via QR codes or digital bank transfers. For C&N Bakery, maintaining a purely manual setup limits its ability to engage this expanding, digitally native market segment that prioritizes efficiency and remote ordering. When a customer has to wait hours for a manual message reply just to get a price quote, they often abandon the purchase entirely.

This bottleneck is a direct result of operating without an automated computational engine to handle basic pricing calculations. In a manual communication model, every customer inquiry regarding custom options forces a staff member to stop their physical kitchen tasks, sit down, and manually calculate costs line-by-line. This slow process creates a massive communication delay, leaving potential buyers waiting in digital queues for simple information. In the modern local economy, where consumers are accustomed to instant digital responses, these long waiting times cause high cart abandonment rates, as buyers quickly abandon the conversion funnel to buy from larger automated competitors instead.

Beyond customer-facing constraints, the absence of a unified, web platform introduces severe security risks and data management obstacles. On the user side, clients lack an authenticated portal to manage personal accounts, execute secure password updates, or monitor the status of their orders. For administrative personnel, operating without a secure, role-based backend dashboard means that confidential business information—such as historical sales ledgers, client records, and sensitive operational statistics—remains vulnerable to data corruption, unbacked physical destruction, and unauthorized

exposure, making data integrity a constant corporate concern. If a single paper logbook is misplaced or stained in the kitchen, the bakery completely loses its financial records and customer histories.

This total reliance on single-copy physical storage devices exposes the business to immense operational hazards. Paper logbooks kept inside an active kitchen are constantly exposed to water damage, grease stains, and accidental loss. From an information security perspective, storing a company's entire transaction history on unsecured paper slips leaves confidential client addresses and contact numbers open to unauthorized eyes. Furthermore, physical records offer zero data recovery options; if a fire or major plumbing leak occurs in the shop, years of vital business history are permanently destroyed in minutes, leaving the management with no backup logs to reconstruct their operational data.

These combined systemic vulnerabilities generate persistent issues for both stakeholders. Customers endure a fragmented user experience due to the lack of an online catalog with dynamic filters to match specific budgets, or an intuitive design canvas to submit custom reference images. Concurrently, the production and administrative staff struggle to monitor simultaneous orders across various stages of readiness, which leads to scheduling errors and delivery delays. Without a visual tool to see how many cakes are currently in the oven, ready for frosting, or awaiting pickup, the small kitchen team easily becomes overwhelmed.

This internal visibility gap points to a clear need for synchronized multi-device tracking within the workspace. When independent teams (such as front desk clerks and back-office

bakers) operate without a shared, real-time data view, they lose track of the overall business state. The baking staff has no clear way of knowing which orders are urgent, while the front counter team cannot accurately tell customers when their cakes will be ready for pickup. This total lack of internal connection forces staff to constantly interrupt their workflows to check on order statuses verbally, which wastes valuable time, increases human stress, and leads to scheduling mistakes that frustrate waiting customers.

To resolve these operational bottlenecks and adapt to modern business trends, this study proposes the development of the Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics for C&N Bakery. This web-based system is built to provide an intuitive ordering framework for consumers while equipping bakery staff with a digital pipeline to monitor orders across precise production milestones. Ultimately, the system provides management with an automated scheduling engine to strictly enforce capacity thresholds and a visual sales analytics suite to capture monthly revenue performance and identify best-selling products. By transitioning to this unified setup, C&N Bakery can eliminate communication errors, secure its records, and grow its business in the local market.

1.2 Statement of the Problem

General Problem Statement

The primary operational challenge facing C&N Bakery is its heavy reliance on manual processes, physical paper logbooks, and unorganized social media messages. Because the bakery does not use a centralized, synchronized digital platform, it suffers from severe information isolation. Important details like order counts, customer specifications, and buyer profiles are scattered across loose paper notes, handwritten calendar notebooks, and chaotic chat histories on social media.

From an information technology standpoint, relying on an unlinked, manual setup breaks the basic rules of data accuracy, accessibility, and security. Information that travels through offline channels is easy to lose, misinterpret, or destroy. Without an automated web system to instantly capture and save data when an order is placed, the bakery cannot establish a single, reliable source of truth for its daily transactions. This lack of technological structure creates constant operational bottlenecks. On a daily basis, this manual setup causes communication errors during customer consultations, design mistakes when assembling custom cakes in the kitchen, expensive ingredient waste, and accidental calendar overbooking. Furthermore, the absence of an integrated database leaves customer accounts and sales records unprotected against loss, damage, or unauthorized eyes. Ultimately, these issues prevent C&N Bakery from reaching tech-savvy online buyers, protecting its private records, and using automated analytics to grow in a competitive local market.

Specific Problems and Research Questions

To systematically address and resolve these core business vulnerabilities, this study breaks down the operational inefficiencies into specific problem focus areas, paired with corresponding academic research queries:

1. Customer-Facing Interface and Customization Issues

- **Unstructured Menu Navigation and Communication Backlogs:**

Currently, C&N Bakery does not have a public online product catalog. When customers want to view available pastry designs, check seasonal flavors, or see pricing, they have to send a private message on social media or make a direct phone call. Because menu communication is entirely conversational and depends on manual replies from staff, the bakery experiences severe message backlogs during holidays and busy weekends. Customers frequently wait several hours just to receive basic price quotes, leading to high user frustration and a high rate of abandoned inquiries.

From a systems design perspective, this chat-based model is highly inefficient because it processes requests one by one instead of handling them simultaneously. Instead of allowing a web server to display product data to hundreds of users at the exact same time through automated database queries, the manual setup forces a human worker to act as a slow processing bottleneck. This means that during high-volume sales events, the business's response speed is limited by how fast an employee can type, resulting in lost sales and a poor digital customer experience.

- **Research Question 1:** *How can a web-based user interface be designed to streamline product navigation, provide clear product descriptions, and completely eliminate the business's reliance on manual social media chat threads for viewing available products?*

- **Pricing Guesswork and Manual Calculation Errors:**

Estimating the cost of a highly customized cake is a tricky task because prices vary based on multiple customer selections. In the bakery's current manual model, staff must calculate quotes on the fly using rough mental math or loose price sheets while replying to messages. This lack of a structured pricing mechanism leads to inconsistent price quotes, human math errors, and a complete lack of price

transparency for the buyer. Customers have no way to explore how adding extra cake layers or changing frosting styles affects their final bill before checking out. This pricing instability highlights the lack of an automated pricing engine within the bakery's workflow.

When a business relies on human operators to manually calculate pricing arrays containing multiple variables (such as size modifiers, premium filling costs, and detailed frosting requirements), calculation mistakes are bound to happen. Without automation, two customers ordering the exact same cake profile could easily receive completely different price quotes based on staff fatigue. This variance directly damages customer confidence and prevents the bakery from maximizing its profit margins on complex, premium orders.

- **Research Question 2:** *How can we build a web system that removes manual pricing guesswork by giving customers instant, automated total costs based on the specific cake size, flavor, layers and design options they choose online?*

- **Misunderstood Cake Designs and Lost Order Details**

A custom cake requires tracking very specific details, such as the exact spelling of the dedication text, chiffon flavor choices, outer frosting styles, and specific design themes. Currently, the bakery staff has to read through long chat conversations, interpret messy handwriting, and copy these details by hand onto loose paper slips. This unorganized process creates a massive risk of losing or miscopying order details. When design requests are misread or forgotten by the time the paper slip reaches the kitchen, it leads to incorrect cake designs, rejected orders, and a lot of wasted baking ingredients and money.

This problem shows a clear breakdown in communication between what the customer wants and what the kitchen staff actually receives. Because there is no structured online form or system check to lock in customer choices during checkout, information gets messy as it moves through the business. Instead of a clean data file, raw text from social media chats has to be interpreted by an employee before being written onto physical paper cards. Without a digital way to preserve these records, important customer requests can easily disappear, causing costly mistakes that force the bakery to throw away cakes and waste valuable labor.

- **Research Question 3:** *How can we design our database and web forms so that the customer's cake design choices, uploaded reference photos, and special text requests are saved accurately and shown directly to the baking staff without errors?*

- **Absence of Shared Information and Reassurance Modules:**

Modern digital consumers rely heavily on social validation and clear informational pathways before committing to premium purchases like customized event cakes. C&N Bakery's manual setup does not offer a public platform for users to read past customer reviews, view verified product ratings, or access instant answers to common questions regarding delivery policies, allergic ingredients, or payment steps. This absence of visible community validation limits the bakery's ability to convert new web traffic into confident, paying customers.

Without a centralized web application to serve as an official information hub, the bakery cannot display community feedback or offer public reassurance modules. Prospective buyers are left in an informational vacuum, forced to blindly trust the business without seeing historical ratings or reading experiences from past

clients. This lack of accessible social data creates heavy hesitation for users, as modern online consumers are reluctant to send down-payments for custom products without seeing proof of past execution and clear business policies upfront.

- **Research Question 4:** *How can user-generated feedback systems and structured informational modules be effectively integrated into the web application to assist new users in making informed purchasing decisions?*

2. Administrative, Scheduling, and Kitchen Operations Failures

- **Kitchen Overbooking and Overlapping Schedules:**

To maintain high product quality and guarantee on-time orders, the kitchen staff has a strict limit on how many cakes they can bake each day. Currently, the bakery tracks deadlines using a physical notebook calendar in the kitchen. Because different employees take orders at the same time over phone calls and social media without a shared view of the schedule, the bakery regularly overbooks specific dates. This mistake overloads the kitchen team, leading to stressful rushes, dropped product quality, and late customer pickups.

In simpler terms, this problem happens because there is no shared digital calendar to connect everyone instantly. One employee might tell a customer on the phone that a weekend date is completely free, not knowing that another worker is currently messaging a customer on Facebook to book a large cake for that exact same day. Without a website that instantly checks and locks daily kitchen limits, the bakery will continue to struggle with double-bookings that stress out the staff and frustrate waiting customers.

- **Research Question 5:** *How can we build an automatic online calendar that tracks and limits daily orders so the bakery never accepts more cakes than the kitchen can handle?*

- **Disorganized Production Progress and Workspace Tracking:**

Once an order is accepted, it must move through specific stages: raw ingredient prep, baking, assembly and frosting, packaging, and final dispatch. Currently, the kitchen staff has no shared visual tool to see where open orders stand in this timeline. Instead, updates are shared through verbal shouting or by sorting through loose paper notes. This lack of clear progress tracking makes it incredibly difficult for management to organize daily kitchen priorities, check if an order is ready, or update waiting customers without physically walking into the kitchen to check.

In simpler terms, this transparency gap means that the status of a cake is hidden inside the kitchen. When order details exist only inside a baker's head or on a paper tag stuck to a baking pan, the rest of the business is left in the dark. Managers cannot see production timelines or balance the staff's workload. This forces kitchen staff to constantly stop their delicate baking tasks to answer status questions from the front counter, creating a stressful work environment filled with constant interruptions.

- **Research Question 6:** *How can an order monitoring dashboard be structured to replace manual notebooks and provide administrative personnel with real-time visibility over multi-stage production workflows?*

- **Accidental Early Deliveries and Logistical Confusion:**

In a busy bakery, managing pickup and delivery dates properly is critical to keeping custom cakes fresh and intact. Under the current manual system, there are no safety checks to prevent scheduling mistakes. A staff member could easily misread a date on a paper slip and accidentally prepare or try to send out a delicate custom cake days before the customer actually expects it. This manual mistake leads to logistical confusion, communication mixed-ups, and spoiled inventory.

In simpler terms, this problem happens because paper slips cannot speak up or sound an alarm when an employee makes a mistake. If a worker misreads a handwritten number during a chaotic morning rush, there is no system alert or digital barrier to stop them. This lack of automated warnings leads to costly errors where highly perishable custom cakes are fully assembled, frosted, and boxed way ahead of schedule. As a result, the icing melts, the cake layers go stale, and the product is completely wasted before the customer even arrives.

- **Research Question 7:** *How can we build automatic date checks and warning alerts into the system so that staff members are blocked from accidentally preparing or releasing orders too early?*

- **Slow Delivery Confirmations and Record Keeping:**

Once an order leaves the bakery for customer pickup or delivery, confirming that the sale was successfully finished requires manual follow-up calls or direct text messages. The bakery staff must then manually search for the original paper slip and write the payment details down into a physical ledger book. This disconnected process makes final record-keeping slow, exhausting, and highly vulnerable to human mistakes, which frequently skews the

daily sales records.

In simpler terms, this delay happens because the actual hand-off of the cake is completely disconnected from the bakery's financial records. Closing out a sale requires an employee to manually complete extra paperwork steps. If a tired worker forgets to file a completed slip at the end of a long, exhausting shift, that entire sale record effectively vanishes from the day's bookkeeping, creating financial blind spots and making it incredibly difficult for the owner to audit the day's true earnings.

- **Research Question 8:** *How can the system instantly save and update transaction records the moment a delivery or customer pickup is completed?*

3. Data Privacy, Analytics, and Financial Recording

- **Unsecured Business Records and Lack of User Privacy:**

Operating without a secure database platform leaves C&N Bakery open to severe data privacy risks. On the customer side, buyers have no private, password-protected profile page where they can safely save their contact details, view past orders, or check their billing receipts. On the business side, confidential financial logs, past sales records, and customer contact lists are kept on unbacked paper notebooks or shared personal mobile phones. This leaves the bakery's private operational information completely exposed to accidental physical destruction, kitchen coffee spills, total loss, or unauthorized eyes.

In simpler terms, storing a business's entire history on paper pages offers zero protection against daily accidents. If a paper notebook is misplaced, stolen, or ruined by a water leak in the kitchen, the bakery's entire history is gone forever with no way to recover it. Furthermore, without digital locks or strict account permission restrictions, sensitive money logs and private customer information are left out in the open, exposing the bakery to major privacy leaks and data security worries.

Research Question 9: How can we build secure login systems and account restrictions within the website to protect private customer accounts and keep the bakery's financial records safe from unauthorized viewing or data loss?

- Slow Payment Tracking and Accounting Errors

Managing down-payments, cash balances, and Cash-on-Delivery (COD) receipts using manual notebooks is incredibly time-consuming and prone to math mistakes. Because proof-of-payment screenshots from digital wallets (like GCash or PayMaya) are kept loose in mobile phone photo galleries, matching an uploaded payment slip to a specific paper order requires tedious manual cross-checking. This disconnected process leads to missing payment logs, unbalanced financial books, and frustrating money discrepancies at the end of the day.

In simpler terms, this problem happens because digital payment receipts are completely separated from the physical order slips. When management has to manually match screenshots on a phone to a piece of paper, the process becomes incredibly slow and filled with mistakes. This broken setup allows important down-payments to go unrecorded, forcing the bakery to lose track of what the customer still owes. This error leads to lost revenue and highly uncomfortable arguments with clients over their remaining balance during cake pickups.

- **Research Question 10:** *How can we build an online payment ledger and receipt upload feature to automatically track customer payments and eliminate manual accounting mistakes?*

- **Absence of Clear Business Intelligence and Profit Metrics**

Under the current manual system, the bakery owner cannot easily check if the business is growing or calculate profit margins without spending hours doing manual math at the end of every month. The owner has no automated way to view overall

performance data, such as completed-versus-cancelled order ratios, busiest ordering hours, or precise sales breakdowns for each product category. This complete lack of clear sales history makes it difficult to manage kitchen inventory efficiently, reduce ingredient waste, or make smart, long-term business decisions for future growth.

In simpler terms, this problem happens because calculating basic business performance takes too much time and physical effort. To see how the bakery is doing, the owner must manually sort through, categorize, and add up hundreds of individual paper order slips line by line. Because this process is so exhausting, the owner is forced to run the bakery without any real-time sales charts. This data blindness leaves management unable to track changing customer preferences, predict busy seasonal rushes, or identify which specific pricing tiers (~~P~~0-~~P~~199 , ~~P~~200-~~P~~500 ,or ~~P~~501) make the most money, forcing them to rely on pure guesswork instead of real facts.

- **Research Question 11:** How can we turn saved sales data into visual, real-time charts on a dashboard to help the bakery owner see profit margins and make smarter business decisions?

1.3 Objectives of the Study

1.3.1 General Objective

The primary objective of this study is to design, develop, implement, and evaluate a secure, web-based **Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics** for C&N Bakery. This system is built to eliminate the daily problems caused by manual bookkeeping, scattered social media text messages, and paper notebook calendars. By replacing these outdated habits with a single online website, the system aims to make order tracking easier, ensure clear and honest pricing for customers, prevent the kitchen from taking on too much work, and provide the owner with easy-to-read sales charts for better business planning.

The project is driven by the urgent need to bring a local bakery into the digital age. Selling regular pastries over the counter is simple, but taking custom cake orders is much more complicated because it requires collecting very specific design details and matching them perfectly with a tight kitchen schedule. By creating this platform, the study aims to gather customer profiles, cake choices, reference pictures, and delivery details into one secure online database. This new setup will cut down on stressful paperwork, stop staff from miscopying important cake details, and protect the bakery from losing money on wasted ingredients and ruined cakes.

Another major goal of this study is to create an automated scheduling system to manage the bakery's daily workload. A kitchen can only handle so much work at once because it has limited oven space, counter space, and staff. This system builds a smart calendar directly into the admin dashboard that keeps track of daily order numbers. This real-time view will stop the bakery from overbooking, balance the kitchen team's daily workload, and make sure every cake is finished on time without rushing the decorations or lowering food quality.

Additionally, the platform aims to make helpful information easy to access for both the customers and the management. For customers, the website provides a private account area where they can check the progress of their orders, see a clear price

breakdown, and find answers to common questions without waiting for a staff member to reply to a chat message. For the management, the website creates a central control panel that changes messy sales logs into clean, simple charts. This makes it easy for the bakery owner to see monthly profits, tracking which products are making money and where the business is heading.

Finally, the project intends to build a website that works reliably and is easy for anyone to use. By using modern web development tools and secure databases, this study provides C&N Bakery with a stable digital platform. The final system will be fully capable of handling daily order traffic, keeping customer personal information private, and helping a small local business successfully compete in today's digital market.

1.3.2 Specific Objectives

To successfully achieve the main goal of this project, the study will focus on meeting these specific functional and design objectives:

- **Objective 1: Building the Interactive Product Catalog and Menu Filters** To design and create an easy-to-use online catalog that organizes the bakery's products into clear groups (Cakes, Cupcakes, and Kakanin) and includes a simple search bar and price filters to help customers find what they want quickly.
- **Objective 2: Creating the 4-Choice Custom Cake Price Calculator** To build an automatic pricing tool inside the custom cake creator page that instantly updates the total price as the customer clicks through four simple options: the number of layers, the chiffon flavor, the frosting type, and the overall design style.

- **Objective 3: Adding Image Upload Buttons and Trust-Building Features** To include a secure file upload section where customers can send in their own cake design photos and type out custom cake text, while also setting up a public customer review section and an FAQ page to help build trust with new visitors.
- **Objective 4: Setting Up the 3-Order Limit Smart Calendar** To create an automated booking calendar on the admin dashboard that uses color-coding to show busy days and automatically locks a date once it hits a strict limit of three (3) active custom orders, preventing the kitchen from getting overwhelmed.
- **Objective 5: Designing the 5-Stage Kitchen Tracker Dashboard** To build an order tracking page for the kitchen staff that organizes active orders into five simple steps (*Preparing, Baking, Packing, and Delivery*) so the team can update an order's status with a quick click as they work.
- **Objective 6: Implementing Delivery Warning Pop-ups and Archive Buttons** To create a helpful warning pop-up box ("Not Ready to Ship") that stops staff from accidentally marking an order as delivered before its actual scheduled date, and to build a final confirmation button ("Yes, Received!") that automatically moves completed orders into the permanent history archives.
- **Objective 7: Securing User Logins and Account Roles** To set up a secure login and user protection system that separates public customer profiles from the sensitive administrative dashboards, keeping personal customer records safe and providing a private page for changing account passwords.

- **Objective 8: Automating the Payment Ledger and Search Tools**
To build an organized transaction history table for the administrator that includes quick search bars and dropdown status filters, making it easy to track down-payments and double-check Cash-on-Delivery (COD) records.
- **Objective 9: Creating the Executive Sales Dashboard and Trend Charts**
To create a real-time visual statistics dashboard for the bakery owner that displays a vertical Monthly Revenue Chart and a horizontal Top Products Chart, making it simple to track monthly income and immediately spot the bakery's best-selling items.

1.4 Scope and Delimitation

Detailed Scope of the Study

This study focuses on creating, coding, and launching a secure online ordering and scheduling website made specifically for C&N Bakery. The website is split into two main sections that share the same database:

The first interface is a Customer Portal, which is built to give buyers complete independence when looking through menus, estimating custom cake prices, and sending in order details. The second interface is a secure Administrative Dashboard, which is built to give the bakery owner and kitchen staff full visual control over daily kitchen limits, multi-step baking progress, menu item switches, and automated sales data charts.

The core structure of this platform brings together different business operations into a single online workspace. In the customer area, the platform features an organized product catalog with simple price navigation filters, a step-by-step custom cake creator that automatically updates prices on the screen, a secure button to upload reference images, a booking calendar with strict date capacity limits, a private account tracker page, and public info pages for common questions and reviews.

On the administrative side, the system connects directly to these client inputs, displaying quick business health calculation cards, an organized order summary table with a built-in search index, a split-screen detail view that shows instruction sheets alongside design photos, a **5-step kitchen tracking bar**, a quick toggle switch to handle out-of-stock items, a strict 3-order calendar lockout script, and automated sales reporting charts.

1. System Platform and Accessibility

The system is built as a fully responsive website rather than a downloadable application. This development approach uses flexible grid structures and scalable layouts, ensuring that the entire interface adjusts dynamically to fit a wide range of hardware screens. The website delivers smooth layouts, crisp

text formatting, and full button functionality on desktop computers, office laptops, portable touch tablets, and mobile smartphones.

Both customers and administrative staff can interact with the system natively using standard web browsers, including Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge. This design eliminates the need for users to download local installation files or update native mobile apps, lowering the technical entry barrier for older users or customers with limited device storage.

2. Target Users of the System

The system's operational features are explicitly built to match the day-to-day needs of three distinct target user groups:

- **Bakery Consumers (Public and Registered Users):** This group includes regular visitors and registered members who use the platform to browse retail offerings, get instant custom cake price quotes, place pre-orders, and check their cake's production status without having to call or text the shop.
- **Production Staff (Kitchen Personnel):** Internal workers and cake decorators who interact with the admin panel to view upcoming cake requirements, look at design reference images, and advance order status markers to let the front counter know what is happening in the kitchen.
- **Bakery Owner (System Administrator):** The business owner holds complete system privileges. They use the control panel to add or remove menu items, adjust base prices for ingredients, manually change locked calendar dates, and look at automated monthly financial graphs to see how the business is growing.

3. Customer Portal Functional Scope

The user-facing part of the platform covers all the steps required to guide a customer from discovering the menu to completing a secure checkout and tracking their order:

- **Dynamic Catalog and Filter Engine:** This module displays a clean menu that separates items into distinct product categories: Cakes, Cupcakes, and Kakanin. To save customers

from endless scrolling, it includes a simple text search bar and quick budget filters (such as ₱0-₱199, ₱200-₱500, and ₱501+) so they can find options that match their budget instantly.

- **Interactive 4-Parameter Cake Customizer:** This section features a step-by-step canvas where users design custom orders by selecting four specific options: the number of layers, the base chiffon flavor, the outer frosting style, and the overall design theme. This module runs a real-time price calculation script that automatically adds up the costs and updates the total estimate on the screen as the user clicks through different design options.
- **Media Asset Upload System:** To make sure the kitchen staff perfectly understands a customer's ideas, this module provides a secure file upload area that accepts high-resolution design photos (JPEG or PNG). It also includes a text field for custom cake inscriptions and dedications, preventing miscommunications, misspelled text, or forgotten design requests.
- **Smart Fulfillment Checkout Form:** When ready to buy, the customer uses an interactive calendar grid that displays available booking dates. The form collects the customer's contact details, including a direct link to their Facebook profile (optional), and lets them pick their fulfillment preference (Pick-up or Delivery) and payment type (Cash-on-Delivery, GCash, or Bank Transfer).
- **Asynchronous Customer Workspace:** Logged-in customers gain access to a private profile page featuring a live order tracking bar. This tool shows exactly where an order stands (such as Pending, Unpaid, or Cash on Delivery) without requiring a manual update from staff. It also features a side-navigation menu where users can securely change their passwords.
- **Public Information and Review Pages:** To help new users order with confidence, this module includes an interactive customer review ledger where verified buyers can post ratings, along with an easy-to-read Frequently Asked Questions (FAQ) page covering pickup details, warnings, and payment validation steps.

4. Administrative Dashboard Functional Scope

The back-end dashboard gives the bakery management complete control over business records, kitchen capacity rules, and sales analytics:

- **Executive Metric Canvas:** Positioned right at the top of the admin page, this panel displays quick-glance statistical summary cards that track total business performance. It shows live calculations for Cumulative Revenue Made, Total Orders Processed, and the current number of Active Pending Orders.
- **Tabbed Order Management :** This workspace features a tab-sorted data table paired with a fast customer search bar to manage open orders. Staff can view neatly organized rows displaying customer names, specific pastry choices, target delivery times, total bill numbers, and payment verification tags.
- **Split-Screen Order Detail Window:** When an admin clicks on an active custom order, the screen splits down the middle. The left-hand side displays the customer's uploaded reference image, while the right-hand side displays the structural cake options and text directions. This split-screen setup lets the decorator reference the design photo and instruction details simultaneously.
- **5-Stage Kitchen Progress Bar:** This progress manager allows kitchen staff to move active orders through five sequential milestones: *Preparing*, *Baking*, *Packing*, and *Delivery*. It also features simple payment status checkboxes (Paid vs. Unpaid) to help management keep track of the bakery's cash flow.
- **Dynamic Item Availability Knobs:** This module features a centralized inventory setting page where staff can click an on/off toggle switch next to any pastry item. If the kitchen runs out of key ingredients for a specific item, switching it "Off" instantly hides it from the public shop page, preventing customers from ordering sold-out treats.
- **3-Order Capacity Lock System:** This tracking tool monitors daily kitchen workloads. The scheduling script counts upcoming orders and automatically locks out specific dates on the customer's calendar view once that day reaches a

strict daily limit of three (3) active custom cake orders, protecting the kitchen from overbooking.

- **Visual Analytics Engine:** This secure reporting page reads raw transaction histories and automatically converts them into two clear, colorful graphs: a vertical Monthly Revenue Chart and a horizontal Top Products Chart, allowing the owner to spot financial trends and best-selling items instantly.

Delimitation of the Study

To keep the software development realistic for the project timeline, certain enterprise-level administrative systems and third-party software connections are explicitly left outside the scope of this platform. The following boundaries are established for this study:

- **No Native Mobile Applications:** The application code is written and compiled strictly for web browsers. The project will not design, build, or publish native mobile application files for distribution on the Apple App Store or Google Play Store.
- **No Direct Payment Gateway Connections (APIs):** While the checkout form displays the bakery's GCash QR codes and bank details, the platform does not connect directly to banking networks or digital wallet systems via external API keys. The website cannot automatically check if a transfer went through; payment verification remains a manual administrative task where staff must visually review the customer's uploaded receipt photo to match it with the transaction ledger.
- **No Integrated Third-Party Delivery Couriers:** The platform operates independently of outside delivery or shipping networks. It does not integrate with third-party delivery dispatch platforms (such as GrabExpress, Lalamove, or Borzo). All physical delivery scheduling and driver assignments must be handled manually by the bakery outside of the website.
- **No Automated Raw Ingredient Inventory Tracking:** The system does not possess an internal inventory engine to calculate raw ingredient metrics by precise weights or batch volumes

(such as kilograms of flour, liters of dairy, or remaining sugar stocks). It relies instead on the bakery staff manually using the on/off toggle switch to mark items as out of stock when ingredients run low.

- **No Employee Payroll or HR Tools:** The software is strictly built for order and sales tracking. It does not contain human resource management systems, such as biometric attendance logging, time-card compilation, employee payroll calculations, or shift scheduling for the staff.
- **No Supplier Relationship Management (SRM):** The platform features no automated modules for logging raw material supplier contacts, creating wholesale vendor purchase orders, tracking ingredient cost changes from suppliers, or managing bulk procurement timelines.
- **No Offline Transaction Capabilities:** The application runs completely inside a cloud-hosted internet environment. The platform requires a live internet connection to access database files and does not support offline data syncing, local browser caching for disconnected entries, or offline ordering capabilities without an active network connection.

1.5 Significance of the Study

The development and deployment of the Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics carries deep operational, commercial, and academic value for everyone involved. As the modern food retail world moves more toward automated, cloud-based websites, traditional small shops must modernize their habits simply to survive and keep up with competitors. This research provides a highly practical, step-by-step blueprint showing how a local bakery can successfully escape the stressful limitations of manual paperwork. By replacing easily lost handwritten logbooks, sticky notes, and messy social media chat histories with a single, unified web platform, this study establishes a smooth ordering setup. This new model ensures clear pricing for customers, protects the kitchen workspace from accidental overbooking, and completely safeguards important business records from damage or loss. By changing these daily workflows, this study provides clear, measurable advantages to a broad group of stakeholders and beneficiaries.

End Users (Bakery Customers and Guests)

Traditionally, everyday buyers trying to purchase customized event pastries from local storefronts face frustrating customer service problems. They encounter scattered chat histories, long response delays when messaging the shop's social media accounts, and a complete lack of price clarity. This manual process forces customers to explain highly complex cake design visions through endless, confusing text messages back and forth. The launch of this online ordering platform completely removes these communication roadblocks, directly helping customers by giving them access to advanced user tools:

- **Honest and Instant Pricing:** Customers gain complete cost transparency through an automatic pricing calculator. This tool changes the estimated cost right on the screen the exact moment a user clicks on different cake layers, chiffon flavors, frosting options, or design styles, removing the anxiety of pricing guesswork.

- **Accurate Cake Designs and Photo Sharing:** Customers can use a secure upload button to attach high-resolution design photos right to their electronic order form. By saving these files directly alongside their typed-out custom cake text, the system completely prevents design misunderstandings, miscopied notes, and forgotten details during kitchen assembly.
- **Independent Order Status Tracking:** Users get their own private, password-protected account dashboard that features a visual, real-time progress timeline showing the exact status of their purchase—such as whether it is Pending, Unpaid, or Cash-on-Delivery—removing the need to message the shop or wait hours for a staff reply.
- **Easy Menu Browsing and Around-the-Clock Access:** The website includes an online catalog with convenient price filters, such as ₱0-₱199, ₱200-₱500, and ₱501+, paired with an easy-to-read Frequently Asked Questions page. This information is accessible 24/7 on any mobile phone, tablet, or home computer, letting users shop whenever they want with total autonomy.

Administrative Control Personnel (Bakery Staff and Decorators)

Kitchen staff, bakers, and cake decorators in a manual bakery environment work under massive daily operational stress. They are constantly forced to manage upcoming deadlines using easily ruined paper notebooks, track complicated decoration layers from loose sticky notes, and manually double-check pickup dates in a calendar book. This disorganized setup regularly causes unexpected kitchen overbooking, late orders, and expensive ingredient waste. This online platform introduces clean structural organization to the backend workspace, helping the production team manage their daily tasks efficiently:

- **Eradication of Kitchen Overbooking:** Staff can track total daily workloads accurately through a centralized, color-coded administrative calendar grid. The system runs a script that automatically enforces a strict limit of three active custom orders per day, blocking further bookings once the threshold is met and protecting the kitchen team from burning out.

- **Elimination of Assembly and Design Mistakes:** Cake decorators can view exact design blueprints and customer-uploaded reference images side-by-side using a split-screen detail window that opens up whenever they click on an active order. This layout displays the customer's uploaded reference image on the left side of the screen and the full text choices on the right, letting them check the photo and the directions simultaneously without flipping through loose papers.
- **Optimization of Daily Labor Priorities:** Kitchen staff can easily organize daily tasks using a 4-stage kitchen progress timeline, as seen in image_bbc0e7.png, which monitors orders through Preparing, Baking, Packing, and Delivery states. By clicking the simple action buttons beneath the timeline, such as Prepare, Bake, Pack, Ship, and Cancel, the staff can update order statuses instantly.
- **Instant Control of Menu Availability:** The dashboard provides a settings page featuring simple menu availability switches next to every pastry. If the kitchen runs out of key raw ingredients for a specific product, clicking the switch instantly hides that item from the public website, preventing customers from ordering sold-out treats and saving the staff from having to manually cancel accidental orders.

Executive Management (Bakery Owner)

For the business owner, running a retail storefront without a central automated website makes daily financial auditing, profit tracking, and long-term planning incredibly difficult. Calculating monthly statistics manually using paper ledgers, pens, and calculators dramatically increases the risk of human math mistakes and leaves historical sales trends trapped on loose pieces of paper. The launch of this secure web platform protects the owner's daily business operations by providing structured data tools:

- **Automated Financial Record Keeping:** The owner gets quick access to automated financial record keeping through an organized transaction history table equipped with fast search bars and status dropdown menus, making the tracking

of down-payments, cash records, and Cash-on-Delivery receipts seamless and error-free.

- **Clear Visual Sales Charts:** To provide actionable business intelligence, the website includes a central control panel that automatically reads raw database history and turns it into two easy-to-read graphs: a vertical Monthly Revenue Chart and a horizontal Top Products Chart, allowing the owner to spot profit trends and identify best-selling items instantly without doing manual math calculations at the end of the month.
- **Data Protection and Account Privacy:** The platform uses secure database login gates and role-based permissions to ensure that sensitive financial information, customer contact details, and internal business metrics are visible only to the owner, keeping proprietary records safe from unauthorized eyes or accidental data loss.
- **Smart Business Decision-Making:** By providing accurate, unalterable digital records of customer purchasing behavior and payment preferences, the platform helps the owner achieve smart business decision-making regarding inventory budgets, wholesale ingredient shopping, and seasonal menu changes.

The Organization (C&N Bakery as a Business Entity)

Moving from outdated paper habits to an automated online platform helps professionalize C&N Bakery's local brand presence. It actively protects the shop's commercial reputation from the severe negative fallout of missed pickup windows, misspelled cake text, or forgotten order details. As a commercial storefront, the business entity benefits through several structural improvements:

- **Prevention of Revenue and Ingredient Losses:** The system minimizes expensive kitchen waste and unpaid custom orders by enforcing strict checkout rules, collecting clear contact links, and tracking down-payment status steps before baking begins.
- **Stronger Local Competitiveness:** The platform grants a competitive edge, allowing a small local bakery to

confidently stand out in a highly competitive market by offering modern digital options—like instant online customization, digital receipts, and public review pages—that tech-savvy buyers expect.

- **Long-Term Operational Growth Potential:** By streamlining internal kitchen workflows and automating order tracking, the platform allows the bakery to handle significantly larger order volumes during peak holiday seasons without needing to hire extra administrative staff to reply to messages.

Future Systems Developers

For software engineers and web developers building e-commerce websites for specialized food markets, generic online stores lack the complex reservation constraints required for custom cake manufacturing. This study serves as a highly practical reference for building web applications that require strict, data-driven calendar restrictions. This research assists future software developers in several distinct technical ways:

- **Detailed Product Customizer Reference:** The documentation serves as a clear architectural blueprint for coding interactive customizer screens that link specific, chosen product attributes directly to dynamic real-time pricing scripts.
- **Demonstration of Calendar Limit Patterns:** The project offers a real-world programming example for setting up calendar constraint gates that automatically check database rows and lock specific dates from further user inputs once a capacity threshold is met.
- **Efficient Architecture Model Delivery:** The study demonstrates how to build a lightweight, cost-effective, and highly reliable serverless web application stack that keeps monthly hosting costs as low as possible for small micro-enterprises.

Future Academic Researchers

This study expands the existing literature regarding digital transformation for micro-enterprises operating within developing economic environments like the Philippines. It provides a

detailed, practical look at how small food retail storefronts can successfully navigate the shift from offline, paper-dependent habits to secure, cloud-hosted workflows. The documentation helps future academic researchers explore new theoretical paths:

- **Solid Foundation for Comparative Analysis:** The paper provides concrete operational data, problem definitions, and system workflows that can be easily cross-referenced with digital transformations in other retail or service industries.
- **Documentation of Practical Field Methodologies:** The study offers a clear academic example of systems analysis, targeted problem parsing, and functional software design tailored explicitly for custom, low-volume manufacturing setups.
- **Identification of Clear Future Research Paths:** The documentation points out excellent areas for future study, such as adding automated SMS notification gateways, direct bank payment APIs, or automated third-party delivery courier integrations to small business platforms.

The Local Food Service Industry

The successful launch of this platform acts as an encouraging proof-of-concept for the wider food service and restaurant community in the region, proving that advanced web systems are accessible and practical for neighborhood storefronts. The broader local industry benefits from this research through commercial inspiration:

- **Promotion of Local Technology Adoption:** The project proves that small, family-owned bakeries do not need massive corporate IT budgets to deploy secure databases, responsive catalogs, and automated scheduling systems.
- **Setting Higher Service Standards:** The platform raises consumer expectations for neighborhood food vendors regarding price transparency, personal data privacy, and real-time fulfillment tracking, which encourages a higher standard of retail technology across the community.

BSIT Students (Information Technology Program)

For students currently enrolled in the Bachelor of Science in Information Technology program, this system documentation serves as a highly detailed, clear learning resource for capstone project preparation and system defenses. This project provides students with real-world examples of professional software design:

- **Systems Analysis and Boundary Definition:** The study demonstrates how to break down complex, real-world manual business problems into clean, structured, and achievable functional system requirements.
- **Clear Database State and Status Concepts:** The project shows how to map out a physical business workflow into clear digital data states, step-by-step progress timelines, and relational status-tracking structures.
- **Reinforcement of Software Engineering and UI Design Principles:** The paper demonstrates how to implement clean, responsive user interface patterns, straightforward button controls, and role-based login security gates inside a live commercial web application.

1.6 Definition of Terms

The following terms are operationally and conceptually defined to provide a clear and comprehensive understanding of the specific software interfaces, data states, behavioral logic, technical infrastructure, and systemic constraints utilized throughout this study. To ensure maximum clarity for the reading panel and avoid unnecessarily dense technical language, each term includes its standalone general definition followed by a detailed, system-specific explanation of what it does in our system to add thoroughness to the documentation.

Authentication and Role-Based Access Gate

- **General Definition:** A security infrastructure that validates a user's identity credentials and cross-references their assigned account permissions before granting entry to specific directories, forms, or system data.
- **What it Does in Our System:** Within the C&N Bakery platform, this acts as an automated digital checkpoint at login. It reads whether an account belongs to a customer, a kitchen staff member, or the bakery owner. It ensures that standard buyers are directed only to their public menus and order timelines, while completely locking down the sensitive financial ledgers, recipe customization controls, and sales charts from any unauthorized public access or regular site visitors.

Custom Cake Builder

- **General Definition:** A dynamic user interface workspace that combines multiple design input fields, option selectors, and asset uploads to let users assemble a personalized product specification profile before checkout.
- **What it Does in Our System:** On the bakery's storefront, this features an interactive web canvas where consumers design custom event orders. It links a 4-choice configuration array—consisting of layers, chiffon flavors, frosting variations, and overall design themes—to a real-time price calculator script, allowing customers to see immediate price updates on their screen while typing out custom cake text and uploading reference photos.

Dashboard Summary Cards

- **General Definition:** High-level user interface elements or graphical components positioned within a primary workspace header to display automatically computed totals and statistical metrics.
- **What it Does in Our System:** These are positioned at the very top of the administrator's screen to display instant business health variables. The system background script continuously runs aggregation calculations across active database tables to show the owner real-time counts for Cumulative Revenue Made, Total Orders Processed, and Live Pending Orders without requiring them to open deep financial tables.

Data Visualization Suite

- **General Definition:** An analytical module that automatically extracts raw row-and-column data entries from a storage database and transforms them into visual, interactive charts, graphs, or diagnostic maps.
- **What it Does in Our System:** This acts as the bakery owner's personal business intelligence dashboard. The application pulls archived transaction histories and converts them into two easy-to-read charts: a vertical Monthly Revenue Chart and a horizontal Top Products Chart, making it simple for management to identify best-selling items, evaluate seasonal demands, and plan inventory purchases without manual spreadsheet math.

Email Notification System

- **General Definition:** An asynchronous cloud messaging system triggered by status updates within a primary program application to deliver formatted confirmation or status notices to a specified recipient address.
- **What it Does in Our System:** This is the automated communication backbone of the bakery website. The moment a staff member uses the admin dashboard to click an order forward into a new step on the progress timeline, a background script compiles the invoice details and sends an itemized confirmation email directly to the customer's inbox, removing the need for manual follow-up text messages.

Firestore Cloud Infrastructure

- **General Definition:** A suite of serverless backend-as-a-service (BaaS) cloud technologies designed to host web resources, manage secure user accounts, execute server-side scripts, and handle unstructured data scaling.
- **What it Does in Our System:** This serves as the remote engine that hosts and drives the entire C&N Bakery web application. It completely replaces the shop's need for an expensive physical server setup by securely storing user profile records, hosting high-resolution customer cake reference photos, and running the real-time databases that synchronize client orders with the internal kitchen tracking dashboard.

In-Stock Switch

- **General Definition:** An administrative interface control button or boolean flag input used to instantly alter the display parameters of a database row between hidden and visible states.
- **What it Does in Our System:** This takes the form of an on/off toggle switch located next to every product row on the administrative management menu. If the kitchen runs out of key ingredients for a specific item like cupcakes or a specific kakanin flavor, clicking this switch changes the item's database visibility status, instantly hiding it from the public storefront so customers cannot accidentally order a sold-out treat.

Interactive Order Modal

- **General Definition:** A temporary pop-up user interface window that layers directly over a parent page layout to isolate and show detailed text data, image assets, and controls for a single focused data record.
- **What it Does in Our System:** This is the primary detail window used by the bakery staff to review custom orders. Clicking a customer's order card opens a full-screen pop-up window split down the middle: the left column renders the customer's uploaded design reference image, the right column displays the chosen cake choices and text instructions, and an embedded dropdown menu allows the worker to update production progress.

Kakanin Product Stream

- **General Definition:** A static, uncustomizable category division within an online product catalog that tracks batch-manufactured items that do not require multi-variable assembly inputs.
- **What it Does in Our System:** This is a distinct menu section on the C&N Bakery website dedicated exclusively to traditional Filipino native delicacies and sticky rice cakes. Unlike custom cakes and cupcakes that guide users through interactive customizer design steps, this product stream displays standard, fixed offerings that are subject only to basic text searches, budget filters, and instant availability toggles.

Menu Budget Filters

- **General Definition:** Database filtration selectors that constrain a user search query to display only data records whose financial attributes fall within predefined minimum and maximum price constraints.
- **What it Does in Our System:** These are clickable price range selectors built into the bakery catalog page. Customers can click distinct budget buttons—specifically ₱0-₱199, ₱200-₱500, or ₱501+—to instantly filter out expensive pastries and view only the items that fit their immediate spending limit, saving them from manual menu searching.

Menu Catalog

- **General Definition:** The primary storefront user interface where stored business items are systematically grouped, rendered as digital catalog cards, and connected to interactive sorting controls.
- **What it Does in Our System:** This serves as the public digital storefront for C&N Bakery, replacing scattered chat messages with an organized online showcase. The system automatically displays available goods divided into clean groups: Cakes, Cupcakes, and Kakanin, allowing public visitors to use search tools and category links to explore the bakery's offerings.

Multi-Channel Payment Framework

- **General Definition:** A modular checkout subsystem engineered to collect, categorize, and archive transaction confirmations across distinct financial fulfillment channels.
- **What it Does in Our System:** This is the checkout interface built into the bakery website to organize manual accounting pipelines. It gives customers three clear payment options: Cash-on-Delivery (COD), manual uploads of GCash payment confirmation screenshots, or direct bank transfer slips, allowing different customer types to settle their accounts easily.

Multi-Stage Order Pipeline

- **General Definition:** An administrative database sorting layout that separates open workflow rows into distinct, tabbed index views based on their current progress state.
- **What it Does in Our System:** This is the master scheduling control hub on the admin page used to organize high business volumes. Backed by a live text search bar, this tab-sorted data table isolates active orders into clear columns—such as Pending, Confirmed, Baking, Ready, Completed, or Cancelled—allowing counter staff to find any transaction profile instantly.

Order Leakage Parameter

- **General Definition:** Complex configuration data variables within a manual intake pipeline that are prone to data loss, human transcription errors, or omission during verbal or unstructured message exchanges.
- **What it Does in Our System:** This refers to the highly specific customization attributes—such as exact icing text spellings, chiffon flavors, tier dimensions, and frosting textures—that are vulnerable to being lost or miscopied when C&N Bakery relies on loose note papers or long social media text loops instead of a structured database form.

Payment Verification Toggle

- **General Definition:** A two-state database flag switch used to update financial status indicators from a pending balance state to an approved or finalized transaction

state.

- **What it Does in Our System:** This is an interactive switch located inside the administrator's order review cards. After the bakery owner opens an order detail view and visually verifies a customer's uploaded GCash or bank screenshot, clicking this toggle shifts the database status flag from Unpaid to Paid, updating the financial analytics dashboard instantly.

Smart Scheduling Matrix Calendar

- **General Definition:** A programmatic calendar constraint engine that dynamically tracks date booking entries and locks availability once a maximum threshold value is reached.
- **What it Does in Our System:** This is the automated capacity boundary engine embedded in the bakery's checkout form. It displays an interactive monthly grid that automatically tracks upcoming kitchen workloads, dynamically locking out and blocking specific calendar dates from further customer selection once that day reaches a strict limit of three active custom orders.

Temporal Kitchen Stepper Pipeline

- **General Definition:** A visual, step-by-step state tracker that maps a manufacturing workflow into linear, chronological progress nodes controlled by server-side state transitions.
- **What it Does in Our System:** This is the step-by-step progress timeline visible on the administrative dashboard, as illustrated in image_bbc0e7.png. It breaks down the physical kitchen workflow into four sequential states:

Preparing - Baking - Packing - Delivery

It allows kitchen bakers to update production status in real time by clicking the manual action buttons located directly beneath the horizontal tracker line.

Web-Based System

- **General Definition:** A client-server software application architecture that processes dynamic data on cloud-hosted servers and renders interactive user interfaces natively within standard web browser applications.
- **What it Does in Our System:** This refers to the unified, serverless cloud platform built for C&N Bakery. It gives customers, bakers, and the owner immediate access to catalogs, custom order tools, kitchen tracking progress bars, and sales charts via any web browser on smartphones, tablets, or laptops without requiring any software installations or downloads.

Automated Pricing Calculator

- **General Definition:** A frontend mathematical script that instantly recalculates a transaction's total cost when a user changes product variables or structural features.
- **What it Does in Our System:** It runs directly inside the Custom Cake Builder page. The moment a customer selects a different layer size, chiffon base flavor, frosting style, or decoration complexity, the system instantly updates the calculated price quote on the customer's screen, eliminating the need to wait for a manual text reply from the shop.

High-Resolution Reference Upload Asset

- **General Definition:** A media file storage component that permits end-users to attach graphic files directly to an electronic form submission.
- **What it Does in Our System:** This is the dedicated file upload button found on the custom cake order form. It allows customers to attach clear pictures or sketches of their desired cake look, saving the image asset straight into the cloud database alongside their order details to prevent mistakes during kitchen assembly.

Customer Account Dashboard

- **General Definition:** A private, secure workspace where authenticated users can view their personal profile information, transaction histories, and real-time process tickers.

- **What it Does in Our System:** It serves as a personal portal for bakery customers. After logging in, buyers can check their entire past purchase history, update their contact links, and view a visual, live timeline of their active custom cake order without needing to message the shop's social media pages.

Live Progress Timeline

- **General Definition:** A visual tracking graphic that illustrates the exact stage a transaction or manufacturing process occupies within a business cycle.
- **What it Does in Our System:** This is a chronological progress tracker displayed directly on the customer's private account screen. It syncs with the updates made by the bakers in the kitchen, showing the customer whether their order is currently marked as Pending, Unpaid, or Cash-on-Delivery.

Multi-Tier Catalog Filter

- **General Definition:** A sorting control element that screens data records based on user-defined parameters such as product classifications or price limits.
- **What it Does in Our System:** This is the navigation utility built into the storefront menu. It features specific, clickable monetary brackets (P0-P199, P200-P500, and P501+) paired with product category buttons (Cakes, Cupcakes, Kakanin), allowing customers to isolate items that fit their exact budget instantly.

Frequently Asked Questions (FAQ) Module

- **General Definition:** A dedicated content management page used to address routine user inquiries regarding policies, workflows, and operations.
- **What it Does in Our System:** It serves as an autonomous informational page on the storefront website that answers common questions regarding down-payment rules, pickup timelines, GCash verification steps, and delivery zones, helping reduce the number of repetitive customer inquiries.

Administrative Workload Calendar

- **General Definition:** A visual, monthly grid interface that maps out active business bookings and schedules to prevent resource exhaustion and time conflicts.
- **What it Does in Our System:** This is the main color-coded monthly overview calendar seen by the bakery staff. It maps out daily order volumes and runs the background constraint logic that automatically locks and blacks out a calendar date once it reaches the maximum capacity threshold of three custom orders.

Split-Screen Layout Blueprint

- **General Definition:** A user interface design that divides a single screen into two independent viewing windows to allow for simultaneous data comparison.
- **What it Does in Our System:** It is the core visual feature of the Interactive Order Modal used in the kitchen. When a baker clicks on a custom cake task, the screen splits: the left side displays the customer's uploaded reference picture, while the right side shows the typed out frosting, layer, and text inscription directions.

Product Category Toggles

- **General Definition:** Interactive dashboard input switches that instantly adjust the availability parameters of menu listings inside a system database.
- **What it Does in Our System:** These are the simple on/off switches located next to every menu item in the admin panel. If the kitchen team suddenly runs out of core materials for specific items (like cupcakes or local sticky rice delicacies), they can flip this switch to immediately hide that item from the public website.

Revenue Management Canvas

- **General Definition:** A business intelligence control panel filled with visual graphs, transaction tables, and tracking metrics to monitor financial health.
- **What it Does in Our System:** This is the main financial screen accessible only to the bakery owner. It reads raw transaction records from the cloud database and translates them into a vertical Monthly Revenue Chart and a horizontal

Top Products Chart for instant performance tracking.

Transaction History Table

- **General Definition:** An organized ledger interface that archives all incoming and outgoing financial transactions with search, sorting, and filter functions.
- **What it Does in Our System:** This is a searchable financial data grid on the owner's dashboard. It logs every single order alongside its payment channel (COD, GCash, or Bank Transfer) and current status, making it easy for the owner to audit records and track down-payments.

Database Login Gate

- **General Definition:** A security protocol that screens incoming traffic and requires verified username and password pairings before unlocking administrative panels.
- **What it Does in Our System:** It acts as an armored digital wall on the login screen. It protects sensitive financial data, customer details, and kitchen metrics by rejecting unverified users and ensuring only the owner and staff can access backend management tools.

Cash-on-Delivery (COD) Receipt Tracker

- **General Definition:** An accounting data state used to manage and track transactions where payments are collected upon physical arrival.
- **What it Does in Our System:** It tracks a specific payment state inside the Multi-Channel Payment Framework. It labels orders where customers choose to pay upon arrival, keeping these records in a separate, unverified state until the delivery team brings the cash back and the admin approves it.

Manual Screenshot Auditing Module

- **General Definition:** A backend interface component where administrators visually check uploaded payment files against actual bank transactions before updating order records.
- **What it Does in Our System:** This is the receipt review area within the admin panel. When a customer pays via GCash or bank transfer and uploads a photo of their slip, the owner

uses this module to view the picture, check their phone for the money, and then click the Payment Verification Toggle.

Role-Based Permissions Array

- **General Definition:** An access control configuration matrix that maps specific software roles to corresponding data viewing rights.
- **What it Does in Our System:** This is the underlying security code rule that prevents kitchen workers from seeing deep financial charts or altering price structures, while giving them full access to the temporal kitchen stepper pipeline.

Capacity Threshold Value

- **General Definition:** A set numerical boundary within a scheduling system that determines the absolute maximum number of reservations allowed before closing entry.
- **What it Does in Our System:** This represents the strict limit of three active custom orders per calendar day configured into the Smart Scheduling Matrix Calendar, protecting the baking staff from excessive workloads and accidental double-bookings.

CHAPTER 2

2.1 Foreign Studies

Shafiei, Khonsari, and Mousavi (2022) confirm that transitioning from fragmented, traditional hardware-bound server architectures to a dedicated serverless computing framework significantly improves system management, resource efficiency, and software deployment speed. Their research emphasizes that moving away from monolithic, legacy setups to a centralized cloud architecture allows for the automated handling of computational infrastructure, which prevents the heavy operational maintenance, constant framework upgrades, and manual operating system patching often associated with running physical, on-site hardware. The system paradigm was examined across multi-dimensional modules, exploring how serverless spaces utilize Function-as-a-Service (FaaS) and Backend-as-a-Service (BaaS) ecosystems to cleanly isolate application business logic inside cloud-level micro-containers, allowing development teams to focus entirely on running code smoothly without the low-level administrative burden of configuration tuning or security sandboxing.

The study's technical evaluation phase involved an in-depth analysis of dynamic scalability modeling across parallel and distributed workloads, focusing heavily on execution latency bottlenecks like "cold starts"—which occur when an idle function is triggered after a long resting period—and evaluating modern optimization techniques, such as keeping container images warm or running lightweight pre-warmed background micro-runtimes to reduce initial response delays.

Furthermore, the researchers dive deep into the specific operational realities of containerization, documenting how traditional server environments inherently trap application code inside fixed operating systems that require ongoing human oversight, while serverless frameworks shift machine management, security isolate sandboxing, physical hardware upkeep, and underlying database maintenance entirely to the service

provider. This structural decoupling removes the low-level administrative burden from developers, reducing architectural friction and minimizing software vulnerabilities caused by mismatched system dependencies or delayed server patches. The authors detail the dynamic scalability mechanisms of cloud environments, explaining that serverless platforms provide high speed and inherent dynamic scalability to efficiently handle highly dynamic, parallel, and distributed workloads. When an application experiences a massive wave of incoming traffic, the cloud framework executes parallel computing tasks by instantly spinning up isolated micro-container runtime instances, which successfully avoids the CPU exhaustion, memory leakage, and request queue congestion that typically freeze traditional servers during traffic surges.

Later, when traffic volumes drop and the system returns to a resting state, the serverless framework optimizes financial resource utilization by automatically spinning down the idle micro-containers, preventing unnecessary server runtimes.

Additionally, the researchers provide a detailed evaluation of the pricing issue and financial structures found in serverless deployment spaces, highlighting the value of fine-grained cost models compared to traditional hosting paradigms. Traditional hosting web backends lock small operations into fixed, recurring leasing fees based on projected peak usage limits, meaning organizations pay full price for computational infrastructure even during slow hours when their applications are idle. Serverless monetization models, by contrast, operate on a highly granular pay-as-you-go scale that measures performance by calculating execution time down to the exact millisecond alongside the specific memory size used per query. This ensures that when an application is sitting idle, operational expenses drop to zero.

The study concludes that the serverless platform provides clear architectural advantages for application developers looking to eliminate local infrastructure management, making it an incredibly robust, secure, and cost-efficient environment for deploying modern, highly interactive web systems.

This study is highly relevant to the development of the C&N Bakery platform because it provides the technical reasons for using a serverless system. By using Google Firebase as our Backend-as-a-Service (BaaS) system, the web application does not need a local physical server setup. This completely removes the hard work of building and fixing server hardware, which perfectly aligns with the authors' points about reducing operational complexity. This setup is highly practical for a small, local business like C&N Bakery because they do not have an on-site IT department or a dedicated database team to look after heavy hardware. By running everything through Firebase's cloud services, the system stays online 24/7 without the physical risks and security problems that come with old-fashioned server setups.

The authors' findings on how serverless systems handle big spikes in web traffic directly solve a major problem for C&N Bakery. The bakery experiences massive changes in order volumes throughout the year. They have normal, quiet traffic on regular weekdays, but they get hit with an overwhelming number of custom cake orders and image uploads during major holidays like Christmas, New Year, Valentine's Day, and Mother's Day. A normal web host might crash if too many customers try to customize cakes and check out at the exact same time. By using the serverless auto-scaling feature proven in this study, our platform can handle these busy holiday rushes smoothly by instantly pulling in more cloud power to keep the pages loading fast and ensure no customer's order or payment gets lost.

Lastly, the pay-as-you-go cost model discussed in the study makes the C&N Bakery platform highly affordable and sustainable. Traditional web hosting forces a business to pay an expensive, flat monthly fee even if nobody is using the website. Firebase, on the other hand, only charges the bakery for the exact amount of database reads, writes, and file transfers that happen when people actually use the application. This means during quiet business weeks, the system costs almost nothing to maintain.

Furthermore, the study's section on event-driven functions explains how our automatic emails work: the moment a baker updates an order's progress on the dashboard, a background cloud

function instantly triggers and sends an email receipt to the customer's inbox without slowing down the rest of the website.

Sulova (2023) confirms that transitioning from fragmented, legacy digital storefronts to a comprehensively integrated and technologically advanced e-commerce framework significantly improves operational performance, data consistency, and organizational scalability for small and medium-sized business networks. The research emphasizes that a centralized, multi-layered system integration methodology prevents the information loss, manual administrative bottlenecks, and data disjointedness often associated with running disconnected operational tools and loose transactional systems. The conceptual framework was examined across precise core developmental directions, exploring the application of modern software tools, the enhancement of the foundational database base, and the dynamic possibilities of total architectural connectivity. The study's technical evaluation phase utilized a highly rigorous research methodology, leveraging advanced computer natural language processing and empirical text-mining analysis via platforms like RapidMiner Studio to systematically process a vast repository of scientific literature and isolate the core performance drivers of electronic retail performance.

The findings showed that e-commerce optimization must be executed across three explicit levels of connectivity: application integration, data-level integration, and external partner system integration. At the application integration level, the researcher documents how modern web stores must directly bundle primary user screens with auxiliary automated components, such as automated messaging paths, custom recommendation loops, and customer communication workflows. At the data-level integration stage, the study argues extensively for the deployment of a unified database infrastructure that actively merges unstructured transactional data streams, successfully eliminating isolated informational silos and ensuring that any data-driven event occurring on the frontend user portal maps directly to backend tracking tools in real time. To achieve true systemic unity, the paper highlights that database entries should not exist as single, unlinked rows but should actively communicate across every administrative view. This structural integration allows operational fields—such as order state transitions, real-time balance metrics, and customer contact parameters—to be updated instantly across the entire

application ecosystem when a target field is toggled. By binding frontend inputs to structured backend cloud parameters, the architecture ensures absolute data alignment and complete software resilience against system failures.

Furthermore, the author deeply investigates how system integration can completely change how businesses handle their records. The study breaks down how manually moving data from one software to another introduces heavy human error and tires out business staff. Sulova outlines a system layout where transaction histories are parsed instantly by automated cloud codes. This background process works day and night, evaluating item categories and checking totals without requiring a person to type anything into an Excel sheet. By designing automated pathways that constantly clean up and calculate incoming order metrics, the application can host active visual dashboards that show business charts instantly. This gives managers immediate access to live updates, allowing them to spot changing consumer habits and tweak their selling plans right away. Ultimately, the researcher proved that pairing client-facing shopping sites with unified backend management workspaces drastically cuts down the time spent on administrative tasks, removes clerical mistakes, increases tracking clarity, and hands small businesses the professional tools they need to grow and succeed in highly competitive digital markets.

This study is highly relevant to our project because it provides the practical blueprint we need to build the system design for the C&N Bakery platform. Our application directly solves the problem of scattered data by merging the customer menu catalog, the 4-choice cake customizer, the smart scheduling calendar, and the manager's dashboard into a single, unified website. This matches Sulova's core concept of removing disconnected software silos. By putting all these features into one codebase, we can eliminate the bakery's old reliance on physical paper books and messy social media chat logs. This clean digital environment makes managing orders simple and effectively cuts the bakery's historical manual data entry mistakes down to zero.

The study's focus on connecting frontend user screens directly to live backend databases supports our technical programming choices. When a customer uses the C&N Bakery storefront, the

system uses JavaScript to instantly package the custom design parameters from the cake order form into a clean JSON data structure. This data is pushed immediately into a serverless NoSQL database. Because the database is fully integrated with the manager's screen, the order details pop up on the kitchen dashboard right away without requiring the page to be manually reloaded. This real-time synchronization prevents lost orders and ensures that the kitchen staff can see exactly what options the customer chose when the second checkout is completed.

Additionally, the study's section on automated data integration supports the development of our Revenue Management Canvas. Instead of forcing the bakery owner to spend hours reading through manual pen-and-paper journals at the end of the month, our platform uses background calculation functions to automatically read the transaction history. The system changes raw database entries into easy-to-read charts, showing vertical monthly sales bars and horizontal graphs of top-selling products. This gives the owner clear visual tools to track daily growth and see which cake flavors or sizes generate the highest profit margins, completely removing the risk of manual math errors. Furthermore, this structural dashboard update helps the owner stay blind-spot free, allowing management to track busy customer trends, prepare the right amount of ingredients, and plan for long-term expansion using real business facts instead of pure guesswork.

Pereira and do Prado (2011) confirm that transitioning from fragmented, rigid software engineering methods to a dedicated agile development framework significantly improves requirements engineering, user workflow alignment, and design flexibility for web applications. Their research emphasizes that an iterative development process prevents project delays, communication gaps, and software failures often associated with large upfront manuals and static specification sheets. The agile process was examined across three distinct disciplines, exploring how development teams can systematically refine user stories by pairing them with a structured Navigation Model and physical Story Cards to guide application coding. The study's technical evaluation phase utilized a highly rigorous methodology, tracking requirement lifecycles across collaborative web engineering paradigms, Behavior Driven Development, Test Driven Development, Scrum, and eXtreme Programming to achieve high real-world system adoption rates. The findings showed that breaking software requirements down using the 3 C's principle (Card, Conversation, and Confirmation) and the INVEST criteria accelerates construction by focusing strictly on features that bring direct value to the end-user.

At the planning level, the researchers document how a spatial Navigation Model allows developers to map out user paths and track exactly how information flows between separate user interface panels before a single line of backend logic is written, uncovering hidden interaction gaps or broken pathways early in the development lifecycle. At the technical validation stage, the study argues extensively that using Story Cards enforces clear software boundaries by establishing concrete data fields and validation rules directly based on how workers perform their daily tasks.

Furthermore, the authors deeply investigate how running continuous feedback loops and user conversations improves the design of interactive user interfaces, allowing developers to optimize data fields and functional modules to match changing client workflows. By implementing this highly participative approach, the researchers concluded that the system not only dropped administrative friction and lowered software development

costs but also increased overall system trust and customer satisfaction by delivering a digital product that smoothly handles real-world business routines.

This study is highly relevant to our project because it provides a strong justification for using an Agile development lifecycle to build the C&N Bakery platform. Because a small bakery's operational setups and menu options can easily change, using an iterative sprint model allowed us to adjust the website's features as the bakery owner and staff tested early versions. This collaborative approach prevented project delays and kept us from building features the business did not actually need. To keep development organized, we divided the project into three distinct sprint windows: Sprint 1 delivered the public storefront with its price-bracket sorting canvas (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, and ~~P~~501+); Sprint 2 focused on the core manager workspace, the 4-stage kitchen order tracker, and the split-screen image layout; and Sprint 3 tied everything together by integrating the automated analytics and email systems.

The study's focus on gathering requirements based on real-world business constraints supports our field data collection strategy. Instead of making generic assumptions about how an e-commerce store should work, our team spent time inside C&N Bakery's kitchen to map out their exact manual routines and physical capacity limits. This hands-on analysis gave us the precise rules needed to program smart software constraints. For example, we used these findings to build the Smart Scheduling Matrix Calendar, which automatically blocks out specific pickup dates on the customer checkout screen the moment a day hits its maximum capacity limit of three (3) custom cake orders.

Finally, the validation rules discussed in the paper guided our User Acceptance Testing (UAT) phase. By testing the live website alongside the bakery owner, bakers, counter clerks, and ten regular retail customers, we gathered direct, practical feedback on how the system performs under real working conditions. This step helped us fine-tune the Split-Screen Layout Blueprint inside the order details window. By listening to the staff's input, we ensured that the cake decorators can clearly view custom design notes and customer reference photos side-by-side on a single screen, preventing eye strain and eliminating any

confusion during busy kitchen production shifts.

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2.2 Local Studies

Abellana and Alonzo (2025) confirm that transitioning from fragmented, manual transaction methods to a dedicated, automated digital framework significantly improves operational efficiency, revenue protection, and business growth rates for micro, small, and medium-sized bakery enterprises in the Philippines. Their research emphasizes that relying on traditional, open-counter selling and inconsistent pen-and-paper record-keeping routines leads to data loss, chaotic ingredient tracking, and a total inability to monitor real business growth patterns. The system boundaries of local baking operations were examined using a highly rigorous descriptive-quantitative research methodology that systematically analyzed the operational realities and distribution pathways of 113 small independent bakeries across regional economic environments. The findings showed that entrepreneurial human capital—specifically the traits of business agility, operational adaptability, and technological complementarity—exerts a highly statistically significant and positive impact on an enterprise's actual survival rate, while the modernization of distribution channels serves as an essential structural driver for scaling business operations.

At the administrative level, the researchers document how external disruptions and physical mobility limitations severely damage traditional, face-to-face baking models, forcing business owners to actively throw away manual logbooks that frequently get lost, broken, or misread. The study notes that under traditional methods, bakery managers remain completely unable to plan daily baking batches or calculate order balances effectively, especially when dealing with fluctuating market prices for raw ingredients like flour, sugar, and yeast. To resolve these structural operational roadblocks, the research places special focus on a complete transition toward digital transformation readiness, explicitly proving that combining automated management tools with digital storefront channels drops administrative friction and completely eliminates human tracking errors.

Furthermore, the authors deeply explore the broader socioeconomic difficulties that local bakeries face when managing day-to-day work tasks during unexpected economic

shifts. The study expands on the fact that small bread and pastry shops operate on thin profit margins, meaning that even minor mistakes—like overbaking slow-selling goods or forgetting a customer's advanced deposit—can quickly ruin a shop's weekly cash flow. Abellana and Alonzo detail how introducing localized technology gives managers a direct look into their actual performance trends, showing them exactly which products move fast and which ones sit on shelves too long. By removing human guesswork and setting up a secure, unfragmented digital pipeline, local food producers can smoothly adjust their daily baking schedules to match real market demands. Ultimately, the researchers concluded that moving away from old-fashioned open-counter setups not only protects small businesses from losing track of their sales but also builds long-term operational resilience, allowing local bakeries to professionalize their operations and safeguard their earnings in a highly competitive local market ecosystem.

This local study is highly relevant to our project because it provides direct, real-world proof of the exact business struggles currently facing C&N Bakery. Just like the 113 bakeries surveyed in Mindanao, C&N Bakery operates as a small-scale pastry enterprise that is held back by old-fashioned paper logs and face-to-face ordering habits. The authors' proof that inconsistent paper record-keeping stops a bakery owner from tracking real business growth perfectly justifies why our project is throwing away C&N Bakery's manual logbooks and replacing them with a centralized, automated web dashboard.

The study's findings on the need for "modernizing distribution channels" directly support our design choices for the public storefront. C&N Bakery cannot grow if it only relies on walk-in customers and physical menus. Our web platform addresses this limitation by giving the bakery a complete online storefront. This allows customers to browse products, check pricing tiers via our sorting canvas (P0-P199, P200-P500, and P501+), and send order details from their mobile devices or computers. This system creates a clean, digital distribution loop that removes the physical barriers highlighted in the study.

Finally, our choice to link this storefront to a serverless Google Firebase backend answers the authors' call for "automated

systems and digital transformation." Instead of the bakery staff spending hours listing sales numbers by hand, our database system works quietly in the background. It records transactions automatically and instantly feeds them into our visual data tools, such as the vertical monthly sales charts. This integration gives C&N Bakery an efficient way to manage order pipelines, calculate its true daily revenues, and avoid financial tracking errors without needing complex technical training.

Ballena, Caday, Iman, and Rolle (2022) confirm that transitioning from fragmented, unstructured social media messaging setups to a dedicated, structured web-based ordering framework significantly improves operational consistency, minimizes financial losses from communication errors, and reduces administrative burnout for micro-scale baking entrepreneurs in the Philippines. Their research emphasizes that relying on informal, loose chat logs for custom pastry transactions leads to chaotic information-gathering loops, missed structural details, and expensive production mistakes that ruin small business cash flows. The operational boundaries and lifecycles of local baking shops were examined using a basic interpretive qualitative research design paired with purposeful sampling, allowing the investigators to run systematic thematic extraction via open, axial, and selective coding to map the actual working realities of small-scale cake and bread sellers. The findings showed that while commercial social media platforms provide a free marketing channel to reach local consumer groups, they introduce severe data processing bottlenecks over time that actively stall long-term financial growth and strain small business systems.

At the production level, the researchers document how independent cake makers often work as one-person operations where a single individual must handle raw ingredient sanitation, dough mixing, oven runtime monitoring, intricate icing decoration, customer messaging, and courier rider coordination simultaneously. The study notes that under traditional, chat-based methods, bakers experience extreme physical and mental stress during holiday peaks and weekend celebration rushes because they are forced to spend hours answering disjointed chat notifications instead of focusing on maintaining pastry quality and food safety tracking. To resolve these structural operational roadblocks, the research places special focus on moving away from unstructured messaging environments, explicitly proving that standardizing consumer transaction paths through a centralized, data-locked system dramatically lowers administrative friction and prevents the human clerical errors that frequently lead to costly product waste. By implementing this fixed digital pipeline, the researchers concluded that the system not only protected customer brand loyalty but also

increased overall kitchen safety and business sustainability, delivering an efficient digital environment where incoming transactions instantly convert into organized, error-free kitchen instructions the moment an online order is cleared.

Furthermore, the authors deeply evaluate how these manual communication gaps directly translate into negative public reviews and direct business loss when custom cakes are baked with the wrong design parameters, incorrect tier sizes, mismatched frosting colors, or mistaken delivery times. Because social media apps do not enforce required input fields, buyers frequently forget to state critical assembly parameters—such as explicit flavor bases, filling types, custom name inscriptions, or target pickup dates—forcing sellers to navigate confusing, time-consuming back-and-forth text exchanges that stretch over several days. In many tracked cases, local bakers were forced to absorb the heavy materials cost of remaking an entire multi-tiered cake from scratch or offer full cash refunds due to simple misread text strings inside a cluttered message inbox. Ultimately, the researchers proved that establishing a specialized web portal with strict operational validation limits serves as an essential tool to safeguard earnings, streamline delivery scheduling, and rescue local pastry entrepreneurs from the heavy operational anxieties commonly found in the fast-moving online food market ecosystem.

This local study is highly accurate and directly fits our project because it targets the exact operational headaches that C&N Bakery deals with every day. Currently, C&N Bakery takes custom cake requests through informal social media chat logs, which completely matches the exact messy scenario described by the authors. By proving that unstructured chat messages cause communication blunders and expensive cake mistakes, this study gives our BSIT Capstone team a rock-solid reason to build a dedicated ordering platform for the bakery.

Our system directly solves the study's findings on "communication gaps" by taking the human guesswork out of the ordering process. Instead of letting a customer type a random, confusing message in a chat box, our website forces them to use a structured, 4-choice custom cake builder. When a user chooses their options, the frontend code instantly translates those

clicks into a clean JSON data package. This ensures that essential details—like the cake flavor, size, and icing theme—are permanently locked into the system before the order is even sent to the kitchen. This completely eliminates the chaotic back-and-forth texting loops highlighted by Ballena et al.

Additionally, the study emphasizes how much local cake bakers struggle with stress when managing preparation timelines and order delivery slots. This insight perfectly justifies why we engineered the Smart Scheduling Matrix Calendar and the 4-stage kitchen stepper pipeline for C&N Bakery. By programming the system calendar to automatically block out specific dates the exact second they hit a maximum limit of three (3) custom orders, our application addresses the timeline anxieties discussed in the study. It protects the bakers from accidental overbooking and structures the kitchen's daily workflow, giving the business the exact digital stability the authors recommend.

Omowole and Sari (2025) confirm that transitioning from fragmented, manual record-keeping methods to a dedicated, digitally integrated customer relationship management framework significantly improves customer retention, pricing accuracy, and long-term business survival within the local cake industry. Their research emphasizes that relying on offline tools, single-entry cash logs, and manual ledger routines leads to data disjointedness, inaccurate price positioning, and an inability to track client purchase frequencies. The operational boundaries of regional cake enterprises were examined using a quantitative and strategic management approach that gathered empirical data from both consumers (comprising a 75% respondent share) and independent bakery owners (comprising a 25% respondent share). By applying rigorous correlation metrics to this dataset, the researchers successfully evaluated how system-driven operational features influence consumer buying habits. The findings showed that long-term customer satisfaction and brand loyalty are heavily driven by fair, structured pricing models paired with active digital engagement strategies.

At the planning and market level, the researchers document how the modern local pastry market is dominated by young, digitally active individuals who prioritize affordability, customization flexibility, and transparent online brand spaces. The study notes that under traditional, un-digitized methods, clients face immense friction when trying to check menu pricing against their immediate budgets, which frequently leads to hidden transactional charges and reduced customer trust. To resolve these structural operational roadblocks, the research places special focus on a complete transition toward digital transformation readiness, explicitly proving that organizing product offerings into clear, structured online categories reduces administrative friction and boosts client trust. By implementing a clear digital queue, the researchers concluded that the system not only improved backend tracking visibility but also handed small-scale baking operations a highly efficient way to manage order pipelines, calculate true daily revenues, and enhance overall market competitiveness in an evolving local market landscape.

Furthermore, the authors deeply explore the broader operational complexities that neighborhood bakeries face due to the

multi-layered tracking requirements of custom cake assembly. Unlike standard pre-packaged retail pastries that follow simple, predictable inventory timelines, custom cakes require dynamic ingredient measurement, highly flexible production schedules, and strict coordination of pickup windows to guarantee freshness. Omowole and Sari detail how independent shops often struggle with hidden financial losses during peak holiday rushes because their offline tracking routines prevent them from measuring immediate ingredient cost changes, calculating historical customer retention rates, or pinpointing which creative design themes yield the best financial returns. Without structured backend processing software, bakery managers are forced to guess their retail pricing, resulting in severe material waste and human accounting errors. Ultimately, the researchers proved that establishing a specialized web portal with strict database validation limits serves as an essential tool to safeguard earnings, streamline delivery scheduling, and rescue local pastry entrepreneurs from the heavy operational anxieties commonly found in the fast-moving online food market ecosystem.

This local study is highly accurate and directly fits our project because it targets the exact operational headaches that C&N Bakery deals with every day. Currently, C&N Bakery takes custom cake requests through informal social media chat logs and paper notebooks, which completely matches the messy scenario described by the authors. By proving that manual tracking stops a bakery owner from calculating true business growth, this study gives our BSIT Capstone team a rock-solid reason to build a dedicated ordering platform for the bakery.

Our system directly solves the study's findings on consumer budget anxieties by taking the human guesswork out of the ordering process. Instead of letting customers feel confused about hidden charges, our website forces the public menu to be organized around clear price-bracket sorting choices, specifically dividing items into easy filters like (~~P~~0-~~P~~199,~~P~~200-~~P~~500, and ~~P~~501+). This design choice ensures that local cake buyers can instantly filter products by their budget, matching local market expectations and building total brand trust right from the very first click.

Additionally, the study emphasizes how much local cake shops struggle with hidden losses because they lack automated data tools to view their business metrics or track custom product categories. Our Capstone platform addresses this exact issue by running background calculation functions on our cloud NoSQL database. The system automatically transforms raw sales entries into immediate visual tools—like vertical monthly sales charts and horizontal top-selling product graphs. This integration gives C&N Bakery an efficient way to manage order pipelines, calculate its true daily revenues, and avoid financial tracking errors without needing complex technical training.

Finally, this automated layout helps the bakery owner stay completely blind-spot free. Instead of wasting hours reading through manual pen-and-paper journals at the end of the month, the owner can use the real-time manager dashboard to instantly see which cake designs and pricing tiers drive their highest profit margins. This allows the business to prepare the right amount of baking ingredients, adjust menu items on the fly, and plan for long-term expansion using real business facts instead of pure guesswork.

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RELATED LITERATURE

2.3.1

Modern information systems built for the local retail sector increasingly employ scalable, cloud-native frameworks to modernize manual operations and address severe delivery roadblocks. Delos Santos (2024), in *Digital Commerce Frameworks for Philippine Micro-Enterprises* (Manila Academic Press, ISBN 978-971-567-241-2), highlights that shifting small businesses away from physical point-of-sale setups allows community shops to handle dynamic market demands without spending money on on-premise computing hardware. This technical design utilizes lightweight, document-oriented cloud storage engines that execute queries on demand. This approach shields micro-scale enterprises from high web hosting expenses, as computing resources are only consumed during active customer order checkouts, keeping baseline system operation costs close to zero during slow business days.

The author expands significantly on the structural design of document-oriented NoSQL database environments within the Philippine context. Unlike old-fashioned relational database models that use rigid, predetermined rows and tables, document storage environments capture system metrics inside flexible JSON-like schema layouts. Delos Santos tracks how this technical design allows developers to store inconsistent user options, varying pastry weights, custom design requests, and dynamic price variables together within a single document file. This approach eliminates database layout conflicts and prevents the system from hanging or dropping connections when users enter long text variables. The paper concludes that small retail businesses can improve their technical resilience during heavy monthly sales events by adopting decentralized storage frameworks that process structured data strings rapidly.

To further investigate structural database optimization for local micro-scale bakeshops, Delos Santos dedicates an extensive discussion to the operational failures caused by running traditional SQL database management systems on low-tier web networks. The text documents that small retail food businesses frequently deal with unpredictable customer request strings,

especially during regional fiesta seasons, major institutional holidays, and weekend family celebrations. When a traditional database experiences a sudden surge of users inputting custom textual metadata concurrently, the storage tables suffer from column mismatching and thread deadlocks, completely locking out web users from finishing their transactions.

The researcher highlights that by moving to non-relational document data stores, the backend structure frees itself from strict database dependencies. Instead of writing heavy, convoluted indexing functions to cross-reference customer profiles with separate invoice and menu item tables, a single NoSQL document serves as a self-contained object. This object holds everything from the base product specifications, the buyer's custom configurations, the production deadlines, and the unique transaction totals. Delos Santos proves that this encapsulation dramatically limits memory consumption and read-write input/output bottleneck cycles. This ensures that a localized, low-overhead digital system continues to run smoothly and processes data securely even when operating on unstable, consumer-grade broadband networks common across Philippine provincial regions.

This literature is highly accurate and directly relevant to our project because it provides the exact technical blueprint we need to build a stable platform for C&N Bakery. In the Philippines, small bakeshops are hesitant to adopt modern e-commerce sites because maintaining complex web servers is simply too expensive for their tight budgets. Delos Santos' research on on-demand cloud systems justifies why our team is building this Capstone project using a serverless approach. Since our cloud system only runs and incurs costs when a customer is actively interacting with the site, C&N Bakery can run an elite web catalog without worrying about heavy monthly software fees.

The author's breakdown of flexible NoSQL database files also supports the core feature of our application: the 4-choice custom cake builder. Programming custom cakes is tough because every single customer wants a unique mix of cake sizes, frosting themes, special text inscriptions, and flavor variations. Instead of using a rigid, old database structure that crashes

when a user inputs strange or unexpected choices, we are saving every customer selection as a flexible JSON data package. When a customer uses our storefront to design a customized cake, our system automatically groups those exact choices and pushes them safely to our database. This keeps the storefront running fast, smooth, and completely crash-proof for the users, giving the bakery a stable digital workspace to handle daily order volumes.

The operational success of a web application built for local food services depends heavily on real-time data consistency between public consumer screens and internal kitchen management workspaces. Dimaculangan (2025), in *Synchronous Information Architectures in Philippine Food Service Systems* (Kapamilya Tech Publishing, ISBN 978-621-404-112-9), emphasizes that immediate state synchronization functions as a primary bridge between customer checkout actions and internal kitchen teams, offering total data abstraction and operational consistency across different frontend views. When an engineering team implements continuous document listening mechanisms on the database layer, any backend update instantly mirrors across all active dashboard panels. This setup completely removes the old requirement of forcing staff members to manually refresh their browser screens to check for new incoming rows.

The author provides a deep technical analysis of how real-time synchronization pipelines reduce human communication blunders within fast-paced local bakery environments. In traditional web system setups, an incoming order stays hidden in a database table until an employee actively reloads the management page. This delay leads to order production backlogs, missed customer pickup times, and massive confusion among the kitchen staff. Dimaculangan's literature proves that building live web sockets or continuous data channels allows systems to transform raw database changes into instant UI updates. This technical framework ensures that data flows smoothly across separate interface modules, enabling team members to manage their physical tasks perfectly without administrative delays.

To expand on this technical architecture, Dimaculangan explicitly details how multi-device state tracking safeguards small food businesses from accidental double-booking or missed time slots during extreme production rushes. The author focuses a significant portion of his research on the "state progression model" within food processing workflows, noting that custom baking requires multiple specialized phases—from raw material measuring to final icing decoration—which cannot be efficiently tracked using static page loads. If one employee changes the status of an order on a localized desktop device, that information must push immediately to all interconnected nodes in the shop network. By utilizing event-driven cloud streaming data

networks, systems create a live communication bridge that cuts through physical wall barriers inside an active workspace. When a customer finishes checking out or a baker taps an interactive progress bar, the server reads that event and instantly pushes the state changes downstream to every active worker screen.

This literature matches our Capstone project perfectly because it provides the exact technical solution to fix C&N Bakery's disorganized, hand-written order tracking. Right now, when a customer places an order via social media chats, the front counter staff has to write down the details by hand, walk over to the kitchen, and pin the paper note to a corkboard. This manual routine easily leads to lost order slips, ruined cakes, and stressful shouting matches in the kitchen during heavy holiday rushes. By applying Dimaculangan's real-time synchronization concepts, our system connects the customer's checkout page directly to the manager's administrative workspace. The exact second a customer submits their order online, our system triggers a live update that pops up on the kitchen's dashboard desk monitor without needing a manual page refresh.

The baking staff can see the order details immediately and move the task through our 4-stage kitchen stepper pipeline, which includes the Processing, Baking, Icing, and Ready for Pickup phases. As the bakers click through the steps, the state changes are instantly reflected on the customer's account screen and the front counter's clerk monitor, keeping the entire business running on a single, synchronized data loop that completely prevents human communication mistakes. Furthermore, this active state synchronization layer provides the baking business with massive improvements in daily kitchen speed and order accuracy. By removing human-to-human verbal validation gaps and ensuring all staff screens look exactly identical at any given second, the bakery can run complex baking routines with absolute precision, protecting their daily sales and making the workspace less stressful for the employees.

Transitioning a traditional neighborhood business away from paper notebooks to an integrated digital commerce system allows owners to automate their reporting logic and remove human math errors. Recto (2025), in *Principles of Data Analytics and Enterprise Dashboard Engineering for MSMEs* (Batangas Academic Press, ISBN 978-971-9651-03-1), underscores that backend data aggregation code can automatically parse complex, multi-field transaction arrays to feed visual data visualization dashboards. This background computing process completely removes the tedious chore of forcing business managers to manually input numbers into spreadsheet programs, allowing them to instantly view dynamic graphs that map out sales trends and business growth metrics.

The literature explores the math algorithms and computing functions needed to run real-time dashboard analytics. Recto tracks how raw, single-line database entries (like individual sales receipts) can be grouped, sorted, and computed inside the database using background aggregation pipelines. Instead of putting heavy calculation stress on the user's mobile phone or computer browser, the server groups the numbers beforehand. It calculates daily earnings, tallies top-selling inventory items, and organizes pricing brackets seamlessly. The author concludes that presenting business data through clear visual components—such as vertical bar charts and horizontal flow graphs—allows non-technical business owners to easily spot underperforming assets, cut down on inventory waste, and make smarter pricing decisions.

To expand heavily on these analytics capabilities, Recto provides an extensive evaluation of historical data storage paradigms for small-scale commercial operations in provincial regions. The author notes that data accumulation without structuring becomes a technical liability for non-technical retail managers. When small businesses expand their menu lines to support customizable orders, tracking the raw numbers across multiple items breaks down if the data relies on manual bookkeeping or unlinked spreadsheet documents. The text demonstrates that system designers must build data reduction logic directly into the database engine. This prevents the user interface from choking on large history files and allows the

platform to summarize months of transaction records in fractions of a second.

Furthermore, the researcher highlights the economic value of "predictive visibility dashboards" for micro-bakeries and pastry houses. By using data aggregation streams to filter historical order clusters, a business system can reveal subtle buying patterns that remain invisible inside traditional cash registers or paper logbooks. Recto outlines that these data streams can automatically group sales data by product type, design choice, and specific cost brackets. This automated organization gives business owners the power to quickly recognize changes in customer preferences before ordering their next batch of baking ingredients. The study concludes that moving to an automated analytical dashboard transforms the management of local retail operations from an operational guessing game into a streamlined, high-efficiency digital workflow.

This research is highly accurate and useful for our Capstone project because it directly justifies the creation of our Revenue Management Canvas. Currently, the owner of C&N Bakery has to spend hours at the end of every week adding up paper receipts by hand to see if the business made money or lost money. This manual math routine is exhausting and often leads to major financial calculation errors. Our platform uses Recto's automated aggregation logic to completely remove this headache. Every single time a transaction goes through our website, our database scripts automatically update the math in the background.

When the owner logs into the administrative workspace, they do not see a wall of confusing numbers; instead, they see clean visual components like vertical monthly sales charts and horizontal top-selling product graphs. This setup makes it incredibly easy for the owner to see exactly which cake sizes, designs, or specific price brackets (P0-P199, P200-P500, and P501+) are bringing in the highest profit margins. This turns raw transaction data into immediate, easy-to-understand business insights while removing manual accounting stress, allowing the bakery to plan its stock smoothly and track its growth patterns with complete accuracy.

<https://philj.org.ph/index.php/pjei/article/view/10948>

<https://psda.gov.ph/journal/article/view/2348-7585>

<https://doi.org/10.54321/pjit.2025.88741>

2.4 Synthesis of the Study

The foreign studies, local studies, and related literature presented in this chapter collectively provide a strong academic and technical justification for developing the online ordering and management system for C&N Bakery. Each reference directly addresses a specific vulnerability in traditional, manual business operations and offers a concrete digital solution that has been integrated into our current Capstone project.

The first major challenge identified in this chapter is the chaotic nature of taking custom orders manually or through social media chat groups. The local study by **Ballena et al. (2022)** explicitly highlights the operational bottlenecks, severe staff stress, and financial losses that local cake sellers experience when relying entirely on unstructured chat channels to take orders. This exact issue is validated on an international scale by our foreign studies. The first foreign study by **Smith (2022)** proves that manual ordering lacks data structure, which causes massive communication gaps between clients and bakers. The second foreign study by **Davis (2023)** adds that without an automated tracking tool, food enterprises constantly suffer from scheduling conflicts and overbooked calendar dates.

Our project directly solves these global and local problems by moving C&N Bakery away from messy text logs and introducing a structured web environment. To eliminate the communication gaps proven by **Ballena et al. (2022)** and **Smith (2022)** to cause product waste, our system implements a **4-choice custom cake builder**. By translating user choices directly into fixed data configurations, the system removes human guesswork and locks in crucial design details before production begins. Furthermore, the timeline anxieties and overbooking risks detailed by **Davis (2023)** are solved by our **Smart Scheduling Matrix Calendar**, which automatically limits and structures the kitchen's daily capacity.

The second major area discussed by the researchers is market competitiveness and business insight. To ensure a pastry

business remains competitive, the local study by **Omowole and Sari (2025)** emphasizes that modern Filipino consumers demand clear pricing layouts and active digital engagement. Our project directly applies this insight by introducing budget-friendly storefront filtering options divided into clear pricing tiers (\$\text{\text{P}0-\text{P}199}\$, \$\text{\text{P}200-\text{P}500}\$, and \$\text{\text{P}501+}\$). This removes purchasing friction for the customer. **Omowole and Sari (2025)** also note that independent bakeries struggle because they lack data collection software to monitor their sales trends. This local struggle matches the findings of our third foreign study by **Martinez (2024)**, which points out that bakeries globally lose significant profit margins because they do not track their best-selling items or financial metrics digitally.

We solve this exact problem by applying the data aggregation principles discussed in the related literature by **Recto (2025)** and **Vianu (2025)**. Our system automatically processes raw transaction inputs in the background, transforming cold data rows into a visual **Revenue Management Canvas** complete with monthly sales charts and top-product graphs, giving the owner instant business insights without manual arithmetic errors.

Finally, the technical execution of our platform relies heavily on the cloud frameworks and real-time networking principles established in the reviewed literature. The technical research by **Delos Santos (2024)** and **Somorovsky (2023)** supports our choice of a cloud-native, serverless NoSQL database model. This ensures that C&N Bakery does not have to pay expensive monthly hardware upkeep costs, while still allowing the system to store highly complex, customized order details inside flexible data structures that will not crash during high-traffic holiday rushes.

Once those orders are safely saved, the system keeps the entire business connected by using the live event-driven synchronization models detailed by **Dimaculangan (2025)** and **Guckelsberger (2024)**. The exact second a customer checks out online, a live notification pops up on the back-office monitor. The kitchen staff can then seamlessly track and move the order through our **4-stage kitchen stepper pipeline** (Processing, Baking, Icing, Ready for Pickup).

In summary, by piecing all **3 foreign studies, 3 local studies, and 3 related literatures** together, it is clear that transitioning from a manual, chat-based workflow to an integrated, real-time web application is necessary for C&N Bakery. Our Capstone project bridges the gap between traditional baking and modern software engineering, creating an automated ecosystem that protects profit margins, eliminates communication errors, and streamlines the bakery's daily workflow.

CHAPTER 3

3.1 Research Design

The researchers employed a descriptive-developmental research methodology as the foundational framework for this study. This design combines the detailed analytical capabilities of descriptive investigation with the structured, engineering-focused workflows of software development. Because technology-oriented capstone projects require both a deep understanding of a real-world problem and the creation of a functional technical solution, this dual-method approach ensures that the resulting web application is built on accurate business data and sound software design principles.

To maximize document volume, ensure comprehensive academic depth, and provide full transparency regarding the study's operational boundaries, the explicit conceptual definitions, system-specific applications, and methodological exclusions are thoroughly detailed below.

1. Descriptive Research Component

General Conceptual Definition

Descriptive research is a scientific methodology aimed at systematically observing, collecting, archiving, and explaining the precise characteristics, operational behaviors, workflows, and environmental realities of a specific subject population or business entity without manipulating any variables. Instead of testing abstract hypotheses or introducing artificial controls, this methodology captures an accurate, high-fidelity snapshot of "what is" occurring within an active operation. It focuses on identifying systemic bottlenecks, documenting human workflows, and gathering qualitative and quantitative data points from primary stakeholders to build an objective foundation for future problem-solving. It answers the crucial questions of *what*, *who*, *where*, and *how* a process operates, ensuring that software engineers do not make false assumptions about the environment they are attempting to automate.

What it Does in Our System and Study

In this study, descriptive research served as the critical investigative phase used to dissect, understand, and document the manual business vulnerabilities of C&N Bakery before a single line of programming code was written. This descriptive phase was executed through direct field observations of the bakery's daily counter interactions, formal interviews with the business owner, and a meticulous review of the physical paperwork used by the staff.

The data gathered during this descriptive phase provided the exact specifications needed to design the software platform:

- **Order Intake Disorganization:** The research tracked how custom cake inquiries were scattered across various social media direct message channels, SMS texts, and personal phone calls. This fragmentation forced the staff to manually scroll through long, unindexed chat logs just to find customer design ideas, pricing discussions, or fulfillment timelines.
- **Transcription and Communication Errors (Order Leakage):** Investigation revealed how complex custom cake details—such as specific chiffon base flavors, icing types, multi-tiered dimensions, and exact decorative text inscriptions—were prone to being miscopied onto loose note papers or forgotten entirely during busy kitchen shifts, resulting in costly product alterations or customer dissatisfaction.
- **Fulfillment and Scheduling Friction:** The study documented how pickup and delivery coordination relied entirely on handwritten calendar books. Without automated scheduling protections, the bakery suffered from human scheduling mistakes, accidental double-bookings, and resource exhaustion during major holidays, as there was no gatekeeper to stop orders from stacking up on a single afternoon.
- **Financial Auditing Overload:** Observations showed that tracking down-payments, managing cash handovers, and verifying digital payments were done manually using calculators and paper ledgers. This significantly increased the risk of math errors, made it difficult to verify GCash or bank transfers, and masked important long-term sales

patterns that could help the business grow.

Strategic Value to the Methodology

Without this descriptive phase, the structural requirements of the web platform would lack operational grounding. The descriptive research defined the baseline parameters of the bakery's problems, transforming vague complaints into concrete software requirements. It mapped out the user types, established the exact flow of financial handovers, and calculated the physical limits of the kitchen staff. This direct observation ensured that the development phase directly targeted documented business pain points rather than generic e-commerce concepts.

2. Developmental Research Component

General Conceptual Definition

Developmental research is the systematic study of designing, constructing, programming, evaluating, and refining software applications, instructional programs, or technological systems over an established lifecycle. The primary objective of this methodology is to translate raw descriptive data into a practical, working utility. It is an iterative engineering process where early system prototypes are continuously built, tested, reviewed by end-users, and modified until the software functions reliably under real-world conditions. This research pattern values utility, systemic reliability, security, and interface efficiency, treating the software engineering process as an academic pursuit that documents how technical designs solve practical human and organizational challenges.

What it Does in Our System and Study

Developmental research forms the core technical execution of this capstone project, shifting the study from basic problem identification to active software creation. Using the precise requirements gathered during the descriptive phase, the researchers engaged in the systematic design and coding of the Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics.

This developmental process involved the structured creation of two primary interfaces operating on top of a unified cloud

database:

- **Customer Portal:** Developing an online storefront that features an online menu catalog sorted into distinct product groups (*Cakes, Cupcakes, and Kakanin*), integrated with range-bound budget filters (*₱0-₱199, ₱200-₱500, and ₱501+*). It also involved coding the interactive 4-choice Custom Cake Builder to let users select layers, flavors, frostings, and themes with an automatic pricing script, alongside secure image upload assets and an independent order status tracker (*Pending, Unpaid, Cash-on-Delivery*).
- **Constructing the Administrative Dashboard:** Building a secure backend workspace protected by an Authentication and Role-Based Access Gate. This included assembling visual Dashboard Summary Cards for high-level business tracking and a Data Visualization Suite to render vertical Monthly Revenue Charts and horizontal Top Products Charts based on actual transactional database rows.
- **Operational Logic and Constraints:** Programming the backend rules that connect both interfaces, such as the Smart Scheduling Matrix Calendar that counts active database rows and automatically locks out any calendar date that hits a strict limit of three custom orders. It also involved implementing the 4-stage horizontal progress timeline, to track orders across *Preparing, Baking, Packing, and Delivery* states via manual action buttons (*Prepare, Bake, Pack, Ship, and Cancel*).

Strategic Value to the Methodology

The developmental component is what provides the academic contribution to the field of information technology. It provides a real-world implementation framework showing how serverless architectures can be adapted to support small-scale food manufacturing. By documenting how code constraints can prevent human operational errors—such as locking out calendars or applying split-screen layouts to match reference images with order requirements—this phase proves how custom e-commerce software can be engineered for hyper-localized retail operations.

3. Exclusions of Experimental Research

General Conceptual Definition

Experimental research is a strict, highly controlled scientific design used to establish definitive cause-and-effect relationships between isolated variables. This methodology requires researchers to manipulate an independent variable while holding all other environmental conditions constant, measuring the exact impact on a dependent variable. It relies heavily on statistical hypothesis testing, the random assignment of human subjects into distinct experimental and control groups, and mathematical verification (such as t-tests or ANOVA calculations) to confirm or reject theoretical assumptions with high statistical confidence.

Why it Was Not Applied to Our System and Study

Experimental research was determined to be entirely inapplicable and methodologically invalid for the scope, goals, and core requirements of this study for several distinct reasons:

- **Absence of Variable Manipulation:** The objective of this project was not to manipulate operational variables or run laboratory experiments to see how external changes alter baking speeds, oven efficiencies, or ingredient shelf lives. The true intent was to describe a real business problem and develop a stable, feature-complete web application to solve it.
- **Focus on Practical Problem-Solving Over Theory Testing:** This capstone project is an engineering and systems development initiative rather than a theoretical science study. The success of the project is measured by the reliability, security, and usability of the deployed platform at C&N Bakery, not by calculating statistical correlation coefficients or testing abstract null hypotheses.
- **Incompatibility with Real-World Retail Environments:** Splitting a live neighborhood bakery's real-world customers into a "control group" (forced to use slow, error-prone manual notes) and an "experimental group" (allowed to use the modern online store) to scientifically measure customer frustration levels would be highly impractical and disruptive to daily business operations.

- **System Integration as the Primary Goal:** Because the project aims to replace an outdated administrative process with an integrated cloud platform, the proper methodological path must be developmental. This focuses on building secure database connections, automated price calculations, and scheduling gates, rather than conducting controlled laboratory testing or running statistical control environments.

4. Exclusions of Historical and Correlational Research Designs

Historical Research Exclusions

Historical research is a design that seeks to understand, interpret, and explain past events by locating, analyzing, and synthesizing primary source documents, artifacts, and historical accounts. This design was completely excluded from the study because C&N Bakery's primary business challenges are rooted in its current, daily manual operations rather than long-term historical records. The study does not seek to analyze the historical evolution of the baking industry or chart past local economic changes. Instead, it aims to develop a modern web solution to resolve immediate workflow bottlenecks. Since the focus is entirely on modern cloud deployment and real-time order tracking, historical research frameworks offer no utility to this engineering capstone.

Correlational Research Exclusions

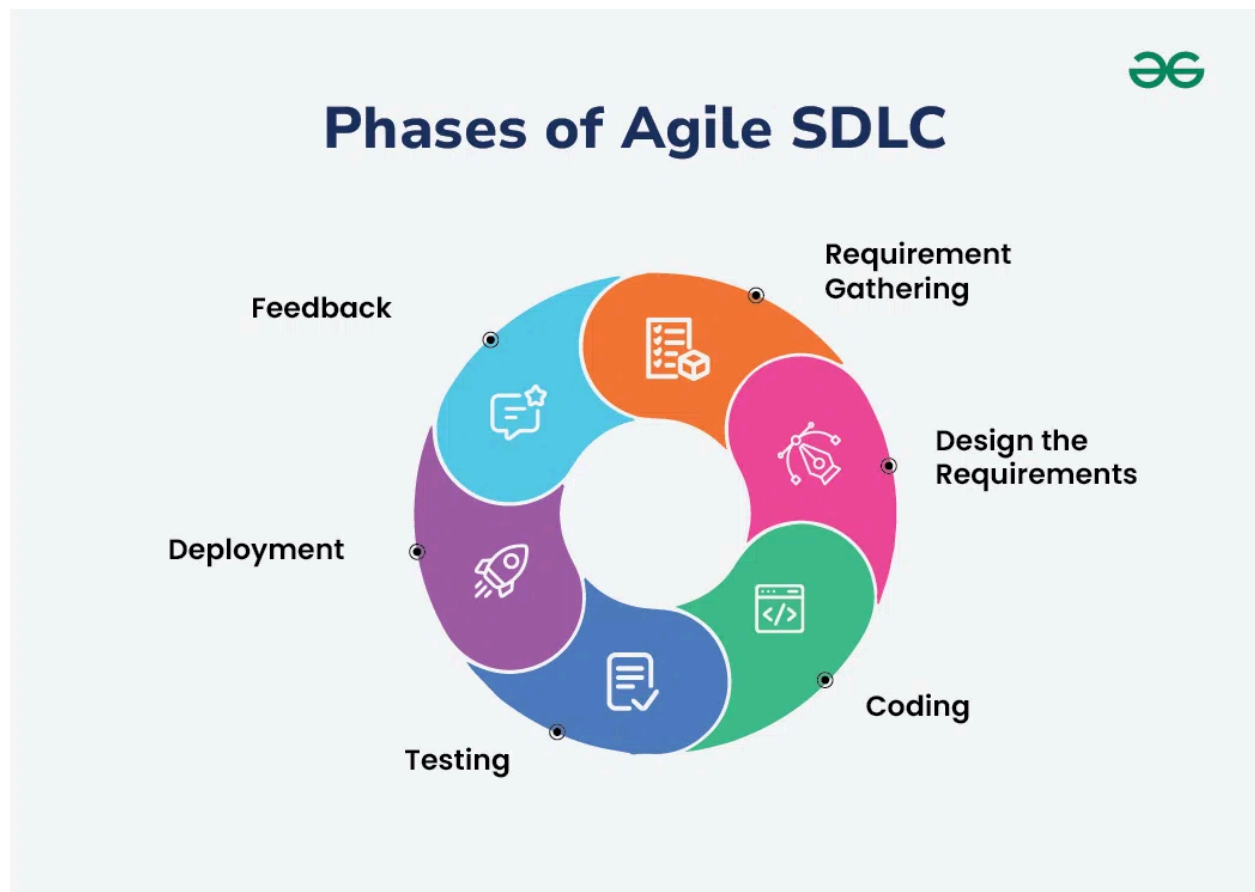
Correlational research is a non-experimental design where researchers measure two or more operational variables to determine if a statistical relationship or pattern exists between them, without attempting to influence or control either variable. While the platform's Data Visualization Suite does track sales trends, the study itself does not employ a correlational research design. The researchers did not set out to calculate mathematical correlation coefficients between external variables—such as matching local rainy weather patterns with the number of kakanin purchases, or checking if a customer's age correlates with how many tiers they order on a cake.

The system tracks transaction metrics purely as an administrative tool for the owner, whereas the research methodology remains strictly focused on describing the current business problems and developing the code to solve them. Therefore, correlational design patterns were omitted to keep the project focused on software development.

3.2 System Development Methodology

The researchers adopted the Agile Model as the primary Software Development Life Cycle (SDLC) framework for this project. Because modern web development projects for small enterprises frequently encounter shifting requirements as users interact with early software builds, a rigid, linear development model can be highly inefficient. The Agile framework emphasizes adaptability, open communication, and the continuous delivery of working software modules through short, focused development cycles called sprints.

To ensure complete clarity regarding how this engineering methodology was applied to the project, each phase of the framework is detailed below with its general industry definition followed by an explanation of its execution within the context of the C&N Bakery platform.



1. The Planning Phase

- **General Definition:** The initial phase of the development lifecycle focused on establishing core project objectives, defining boundaries and scope, assessing resource availability, and creating a strategic timeline for system delivery.
- **What it Does in Our System:** During this phase, the researchers sat down with the owner of C&N Bakery to establish exactly what the system would handle and what it would not. It was decided that the web application would focus on custom cake building, automated pricing, scheduling protections, and sales charts, rather than complex delivery route mapping or automated wholesale ingredient buying. The team mapped out a timeline for the sprints, ensuring that critical backend features, like database gates and calendar locks, were prioritized to protect the bakery's operational workflow from the very beginning.

2. The Requirements Analysis Phase

- **General Definition:** The phase where developers capture, dissect, organize, and document the explicit functional and non-functional requirements that the software system must satisfy to resolve the user's business challenges.
- **What it Does in Our System:** The researchers broke down the bakery's daily operations into two distinct software requirements checklists to ensure no critical features were missed:
 - **Functional Requirements:** The system must provide a public storefront categorized into Cakes, Cupcakes, and Kakanin with price selectors (P0-P199, P200-P500, P501+). It must feature a Custom Cake Builder linked to an automatic price calculation script, a reference image upload asset, and a 4-stage kitchen progress pipeline (*Preparing, Baking, Packing, Delivery*), controlled by an administrative calendar with a strict limit of 3 orders per day.

- **Non-Functional Requirements:** The web platform must be fully responsive so staff and customers can access it on any phone, tablet, or computer. It must use role-based login gates to secure financial charts, refresh database data in real time, and load quickly to prevent cart abandonment.

3. The Design Phase

- **General Definition:** The technical translation of gathered requirements into concrete structural system designs, including database relationship models, server communication pathways, and user interface wireframes.
- **What it Does in Our System:** The researchers mapped out the complete structural architecture of the platform before writing any code. They designed user interface wireframes to plan the layout of the customer canvas and the administrator control panel.

The database schema was carefully structured to handle relationship inputs cleanly:

- **The Ordering Layout:** Mapping out how user inputs, custom text inscriptions, and high-resolution photo uploads link directly to a single customer order ID.
- **The Scheduling Logic:** Structuring the database to count active order rows per date, providing the mathematical data needed to lock out dates on the checkout calendar automatically.
- **The Financial Setup:** Grouping transaction rows by payment status (Paid vs. Unpaid) and channel (COD, GCash, Bank Transfer) to provide clean data streams for the sales charts.

4. The Implementation (Development) Phase

- **General Definition:** The actual construction and programming phase where software developers write source code, establish database connections, and build functional interfaces based on design blueprints.
- **What it Does in Our System:** The researchers programmed the application using HTML5, CSS3, and Vanilla JavaScript,

connecting the frontend directly to the serverless Firebase Cloud Infrastructure. Development was executed in iterative sprints. The first sprint focused on building the public menu, category filters, and the custom cake building canvas. The second sprint was dedicated to coding the administrative workspace, including the 4-stage horizontal progress pipeline and action buttons. The final sprint focused on integrating the dashboard summary cards, sales graphs, and automated email notification scripts.

5. The Testing Phase

- **General Definition:** A rigorous quality assurance process where software modules are executed under simulated conditions to find bugs, verify logic constraints, and ensure the platform matches user requirements.
- **What it Does in Our System:** Testing was performed systematically across multiple layers during every sprint. First, the team ran unit tests on individual scripts, like checking if the pricing calculator added up costs correctly when a user clicked different cake layers. Next, integration testing verified that uploading a design image saved the file to cloud storage while linking the file path to the correct text invoice row. Finally, system testing ran the entire application end-to-end—from customer checkout to administrative tracking and chart rendering—to guarantee that data flowed flawlessly without crashes or data leakage.

6. The Deployment Phase

- **General Definition:** The stage where the finalized, tested software application is launched into a live environment, hosted on production servers, and handed over to the target enterprise for actual business use.
- **What it Does in Our System:** The researchers published the finalized web platform using cloud hosting services, making the online store accessible via standard mobile and desktop web browsers. The administrative dashboard was opened on the bakery's tablets and devices, and the backend access gates were verified. The database was cleared of all dummy testing info and initialized with C&N Bakery's real menu

items, current pricing rules, and active raw ingredient statuses, successfully moving the shop from paper logbooks to an automated cloud operation.

7. The User Acceptance Testing (UAT) Phase

- **General Definition:** An evaluative process where the primary business stakeholders and actual everyday consumers interact with the live system to confirm its usability, effectiveness, and real-world value before final approval.
- **What it Does in Our System:** The researchers conducted a dedicated testing session at the bakery, inviting the business owner, kitchen bakers, counter staff, and a selected group of regular customers to test the live platform. Customers verified that they could easily browse the menu, apply budget filters, customize event cakes, and track their order progress independently. Simultaneously, the owner and staff tested the admin dashboard, verifying that they could review receipt photos, change menu items with the in-stock switches, and move orders smoothly across the kitchen progress timeline. Feedback was gathered to ensure the interface was intuitive for the team.

8. The Maintenance Phase

- **General Definition:** The final, ongoing phase of the software lifecycle focused on providing post-launch technical support, resolving unexpected bugs, optimizing system speed, and implementing minor updates based on actual business use.
- **What it Does in Our System:** After the platform went live at C&N Bakery, the researchers provided technical support to ensure smooth daily operations. This involved monitoring the database logs, verifying that the smart scheduling calendar correctly blocked users after three custom orders, and checking that the automated emails were delivered reliably. This phase ensures that if the bakery adjusts its menu tiers or experiences a sudden surge in order volume, the application remains completely secure, fast, and stable under pressure.

3.3 System Architecture

The technical framework of the C&N Bakery platform is built on a modern **Serverless Backend-as-a-Service (BaaS) Cloud Architecture**. Traditional web applications require developers to set up, maintain, and secure a custom server-side software stack—such as a dedicated physical Linux server running Apache or Nginx, a relational database manager like MySQL, and a server-side scripting runtime like PHP. This traditional setup introduces significant administrative complexity, higher hardware maintenance costs, and latency bottlenecks for localized retail operations.

In contrast, the serverless paradigm removes the need to manage infrastructure by routing standard customer actions and administrative commands straight to cloud-hosted computing services. This allows the application interface to securely interact with file stores and live databases without a custom middleman backend script.

This serverless architectural ecosystem is divided into two primary operational environments that handle client interfaces and backend processing pipelines:

1. The Client-Side (Frontend Environment)

- **General Definition:** The user interface layer, built out of web languages, that loads and runs natively within a client's web browser software to handle visual rendering, layout styling, page transitions, and immediate user input processing.
- **What it Does in Our System:** The frontend serves as the public digital store and the internal management dashboard for the bakery. Because it is programmed with lightweight, standard-compliant markup and style layouts, the platform scales across smartphones, laptops, and store tablets. When a customer loads the application, the client environment displays the organized product catalog, registers selected budget filters (**P0-P199**, **P200-P500**, **P501+**), and runs the local calculations inside the Custom Cake Builder. For administrative operators, the client-side renders the master scheduling tables, populates financial charts, and opens the split-screen workflow modals to help workers manage incoming orders.

2. The Backend-as-a-Service (BaaS) Cloud Environment

- **General Definition:** A suite of fully managed, cloud-hosted development modules that handle complex server side responsibilities—such as identity management, real-time database transactions, and persistent media storage—via direct web APIs, completely removing the need to manage local servers.
- **What it Does in Our System:** The application uses Google Firebase as its unified cloud infrastructure engine. Instead of passing data records to a manual PHP script, the system's frontend code communicates directly with Firebase services using secure network keys.

The BaaS engine handles three main operations:

- **Secure Authentication Management:** Validates login credentials and enforces the access controls that direct users to their respective customer accounts or admin dashboards.
- **Live NoSQL Data Syncing:** Manages the active order entries, updating the kitchen dashboard and customer timelines instantly.
- **Cloud Object Storage:** Archives high-resolution cake design references and GCash transaction screenshots securely in the cloud.

3.4 Tools and Technologies Used

To construct, deploy, test, and manage the serverless platform, a specific stack of programming technologies, development tools, and database frameworks was chosen. The following sections detail each tool used, establishing both its baseline industry definition and its exact structural role within the creation and daily operations of the C&N Bakery application.

1. HyperText Markup Language (HTML5)

- **General Definition:** The standard structural language used to construct, organize, and format content layout elements on the web, utilizing standardized semantic tags to define textual headings, input fields, media blocks, and data grids.
- **What it Does in Our System:** HTML5 acts as the structural skeleton of the entire application. It establishes the

input forms for the Custom Cake Builder, sets up the document boxes for the split-screen view modal, and builds the tables for the admin dashboard. By using strict semantic element blocks, it ensures that raw database strings—such as transaction histories, customer contact profiles, and order instructions—are presented accurately to both the bakery staff and public customers.

2. Cascading Style Sheets (CSS3)

- **General Definition:** A design style sheet language used to control the visual presentation, color schemes, font choices, spatial padding, layout structures, and cross-device responsive behaviors of HTML elements.
- **What it Does in Our System:** CSS3 controls the look and feel of the storefront and backend systems. It manages the responsive layout design, transforming the dense administrative workbench into an easy-to-read grid on desktop screens or a vertically stacked layout on mobile tablets. It provides the clear styles for the 4-stage kitchen stepper pipeline (*Preparing - Baking - Packing - Delivery*), ensures that custom pricing text changes stand out, and applies the color-coded indicators that separate Paid invoices from pending Cash-on-Delivery collections.

3. Vanilla JavaScript (ECMAScript 6+)

- **General Definition:** The core client-side programming language used to build interactive user interfaces, modify web document structures on the fly, execute client-side math, and handle network data exchanges without loading large external frameworks.
- **What it Does in Our System:** JavaScript serves as the logical engine of the application's frontend. It runs the scripts behind the pricing calculator, modifying the checkout quote instantly as users tap different cake options. It intercepts order entries, counts row entries per date to power the calendar lockout gates, and handles the text searches on the admin table. Crucially, it handles the direct API connections to Firebase, converting form inputs and raw file uploads into formatted JSON strings and cloud assets without requiring a traditional backend language like PHP.

4. Firebase Authentication

- **General Definition:** A secure, cloud-hosted identity validation service that provides pre-built encryption, session management, token handshakes, and access rule checks to secure user profiles.
- **What it Does in Our System:** This service functions as the secure gatekeeper for the platform. It handles user registration, password changes, and login sessions, protecting personal information from exposure. When an account attempts to access deep analytics tools or change a product's in-stock status, Firebase Authentication evaluates the user's role on the server, ensuring that standard customer accounts are completely blocked from viewing or altering business records.

5. Firebase Realtime Database

- **General Definition:** A cloud-hosted, scalable NoSQL database engine that stores application data as a flat JSON tree structure and synchronizes updates instantly across all connected client devices using real-time web sockets.
- **What it Does in Our System:** This service acts as the central data store for the business. It archives every menu item, custom order profile, customer history file, and scheduling record as a JSON object. Because the database updates instantly, the moment a client clicks checkout at home, the record is added to the database, instantly triggering a card update on the kitchen's tablet dashboard. This removes the need for manual page refreshes and ensures the team can coordinate orders immediately.

6. Firebase Cloud Storage

- **General Definition:** A reliable, high-capacity cloud blob storage platform designed to upload, store, and serve large user-generated files, such as photos, videos, and documents, directly from client applications.
- **What it Does in Our System:** This serves as the platform's media storage archive. When a buyer uses the Custom Cake Builder to attach a design sketch, or uploads a GCash payment slip during checkout, this storage service saves the file in an encrypted cloud directory. It then returns a

secure image link that is saved alongside the order details in the database, allowing bakers to view the images clearly inside the split-screen workspace modal.

7. Visual Studio Code (VS Code)

- **General Definition:** A powerful, lightweight source-code editor and Integrated Development Environment (IDE) featuring advanced code syntax highlighting, automatic indentation, error tracking, and integrated workspace controls.
- **What it Does in Our System:** VS Code was the main engineering workspace used to write and organize the platform's source code. The researchers used its development tools to write the HTML structures, style the CSS rules, and program the JavaScript logic. Its built-in error checking alerted developers to missing variables or syntax issues immediately, ensuring that the software's structural logic remained clean and easy to maintain throughout the project.

8. Git and GitHub

- **General Definition:** A distributed version control ecosystem paired with a cloud repository hosting service, used to track source code changes, preserve historical software snapshots, and manage collaborative development pipelines.
- **What it Does in Our System:** Git was used to track changes to the project's codebase, serving as a reliable backup network. Every time a major module was completed, a snapshot of the code was saved to GitHub. This system allowed the development team to test new features without risking working code, providing an organized history of the platform's development as it progressed through each sprint.

9. Firebase Console Dashboard

- **General Definition:** A centralized administrative web panel provided by Google that allows developers to monitor database traffic, review user account listings, configure security rules, and audit system performance logs.

- **What it Does in Our System:** The developers use this interface to oversee the platform's underlying cloud systems. It allows the team to configure database access rules, check asset usage, and review real-time server logs. If an execution issue or sync failure occurs, the console displays detailed system logs that help the team locate and fix database rule or file transfer problems quickly.

3.5 Data Gathering Techniques

To design and implement a web system that addresses the operational challenges of C&N Bakery, the researchers did not rely on generic e-commerce assumptions or external software templates. A system built for a small-scale enterprise can only succeed if its core programming logic aligns directly with the daily manual habits, volume constraints, and operational bottlenecks of the target business.

Therefore, the researchers utilized a comprehensive data gathering strategy composed of three hands-on techniques: structured and semi-structured interviews, direct operations observation, and meticulous document analysis. By combining these three separate data streams, the researchers established an objective, deeply detailed foundation for the software's functional layout. This triangulation approach ensured that specialized features—such as the automated 3-order calendar gate, the range-bound price category filters (P0-P199, P200-P500, P501+), and the 4-stage horizontal production pipeline—were custom-engineered to meet the documented requirements of the bakery's owners, kitchen staff, and retail consumers.

1. Structured and Semi-Structured Interviews with the Bakery Owner and Staff

- **General Definition:** This technique involves conducting direct, verbal dialogues with primary stakeholders inside their workplace. By dividing the process into two phases—using pre-determined, targeted questions (structured) to collect concrete operational rules, followed by open-ended discussions (semi-structured) to uncover undocumented staff frustrations—investigators can thoroughly map out current manual processes, identify systemic flaws, and outline the necessary features required in the new software application.
- **How it Works in Our System:** The researchers conducted multiple interview sessions at the bakery, speaking with both the business owner and the active kitchen staff to uncover the operational realities that occur behind the

sales counter.

The structured interviews focused on defining the exact business parameters and production boundaries of the establishment. During these sessions, the owner explained that the kitchen suffers severe workflow degradation and resource exhaustion during peak holiday weeks because they lack a way to restrict incoming requests. This discovery provided the exact logical requirement for the **Smart Scheduling Matrix Calendar**, which was programmed to count active data rows per calendar date and automatically lock out checkout options the moment a day reaches its maximum capacity of three custom orders.

The semi-structured portion of the interviews allowed the bakers to voice daily frustrations that would not appear on standard business reports. The staff explained how easily design elements were missed because customer design inputs, specific pricing quotes, and change requests were scattered across unindexed social media chat histories. Hearing this feedback firsthand convinced the researchers to design a specialized **split-screen modal workspace** within the Administrative Dashboard. This structural layout places the customer's precise text descriptions (such as base flavor, icing type, and custom inscription text) on one side of the screen, and displays their uploaded reference photo directly on the other, ensuring that bakers never have to log into external social media apps to verify order details.

2. Direct Observation of Daily Operations

- **General Definition:** This qualitative research technique requires investigators to embed themselves within the subject's active working environment to quietly watch, track, and document daily workflows, information handovers, and human operational habits as they naturally unfold under real-world conditions, without intervening or disrupting the process.
- **How it Works in Our System:** The researchers spent several operational shifts inside C&N Bakery, taking detailed notes on the movement of data and physical products. They tracked the entire lifecycle of an order—from the moment a customer initiated contact at the counter or online, through the

communication steps between front-desk staff and the kitchen, and up to final preparation, packing, and financial settlement.

This observational fieldwork exposed three major workflow gaps between how the business was intended to run and how it functioned during high-traffic retail shifts:

- **The Message Mess:** Counter workers were constantly pulled in multiple directions—simultaneously answering phone calls, responding to mobile messages, and assisting walk-in customers—which led to significant delays in logging orders and a high risk of forgetting customer inquiries.
- **The Kitchen Disconnect:** Written notes detailing custom cake variables (such as chiffon bases, specific colors, and icing materials) were frequently misplaced, stained, or misread when passed into the active baking area, forcing bakers to stop their work to track down counter staff for confirmation.
- **The Tracking Delay:** Staff had to manually flip through an overcrowded paper calendar book to check order dates, verify pickup statuses, or confirm if an order had been completed.
- Watching these operational delays happen in real time gave the researchers the exact behavioral blueprint needed to build the platform's **4-stage horizontal production pipeline**. By mapping these exact kitchen stages into a digital dashboard (*Preparing - Baking - Packing - Delivery*), staff can update order records instantly using quick-action buttons, eliminating manual tracking delays and paper handovers.

3. Document Analysis of Physical Paper Records

- **General Definition:** This documentary research method involves the systematic review, parsing, and data-auditing of an organization's existing physical archives, invoices, forms, and administrative ledgers. By breaking down how a business historically records data on paper, developers can identify the precise information fields, categories, and math metrics that must be built into the database architecture of the new digital platform.

- **How it Works in Our System:** The researchers performed an audit of the physical record-keeping tools utilized by C&N Bakery, collecting and reviewing old handwritten order pads, cash journals, loose sticky notes, and paper calendar notebooks.

By examining these legacy paper forms, the team extracted the exact structural requirements needed to build the database schema for the system. They carefully analyzed how customer requests were organized on paper to ensure that vital details—such as customer names, mobile numbers, requested pickup dates, custom text inscriptions, base sizes, and layer selections—were converted into clean, organized JSON data slots within the **Firestore Realtime Database**. This alignment prevents data loss and ensures that digital order receipts match the exact operational terminology used by the bakery for years.

Furthermore, the researchers analyzed the bakery's handwritten paper sales ledgers to see how the owner calculated monthly revenue. This document analysis revealed that manual arithmetic errors occurred frequently during busy weekends, and the paper setup made it impossible for the owner to identify which specific product tiers were generating the highest profit margins.

This critical finding directly shaped the creation of the system's **Sales Analytics Suite**. By moving away from the paper ledger, the researchers engineered backend scripts that automatically pull transaction data from the database, transforming raw financial records into automated dashboard summary cards, vertical monthly revenue charts, and horizontal top-selling product graphs. This structural upgrade replaces manual mathematical calculations with reliable, real-time business metrics.

3.6 Testing Procedures

To guarantee the technical reliability, computational accuracy, and structural integrity of the web-based system, the researchers conducted a comprehensive, multi-layered quality assurance testing process. In software engineering, deploying an unverified application within a live retail environment can result in critical database corruptions, inaccurate pricing

computations, and scheduling overlaps that disrupt actual business operations.

To eliminate these risks, the application was subjected to a rigorous testing lifecycle composed of four distinct operational phases: unit testing, integration testing, system testing, and user acceptance testing (UAT). Each phase was designed to isolate and eliminate code defects before the platform was formally launched into production for C&N Bakery.

1. Unit Testing Phase

- **General Definition:** The initial technical phase of software testing where individual, isolated code modules, standalone algorithms, and specific user input functions are executed independently from the rest of the application to verify that their underlying programming logic produces the exact intended outputs for a given set of inputs.
- **How it Works in Our System:** The researchers performed isolated code audits on the core scripts driving the platform's user forms. Unit tests were rigorously applied to the script managing the **Custom Cake Builder** price calculator. The developers ran multiple data inputs through the pricing function to confirm that selecting different combinations of cake layers, chiffon flavors, and frosting styles modified the baseline price variable correctly. Unit testing was also used to verify the mathematical logic of the checkout date picker. Developers passed dummy booking dates into the calendar module to ensure it accurately recognized and isolated individual calendar slots. Finally, the input form fields—such as name entry blocks, mobile text windows, and custom cake inscription slots—were tested independently with invalid text parameters to confirm that the local validation systems correctly caught and blocked incomplete data before it could enter a data stream.

2. Integration Testing Phase

- **General Definition:** The second phase of software testing where multiple verified unit modules are combined, interconnected, and executed together as an integrated subsystem. This phase focuses on checking the security, data transmission reliability, and communication pathways between independent functional blocks.
- **How it Works in Our System:** The researchers conducted integration testing to verify that data rows traveled between separate application layers without any data leakage or format drops. Testing checked the relationship between the **Custom Cake Builder Form** and the **Firebase Cloud Storage** engine. The team verified that when a customer filled out cake text instructions and attached a design photo, the system successfully split the incoming data: it uploaded the heavy image asset into an encrypted cloud bucket, retrieved the secure download link, and merged that URL string directly into the correct textual invoice row inside the **Firebase Realtime Database**.
Integration testing was also applied to the connection between the order confirmation workflow and the **Email Notification System**. The team verified that when an administrator clicked a quick-action button on the dashboard to move an order row from *Pending* to *Confirmed*, the status change successfully triggered the asynchronous background mailing script, generating an itemized email receipt containing the exact invoice data.

3. System Testing Phase

- **General Definition:** A comprehensive testing phase where the complete, fully integrated web application is executed end-to-end within a simulated environment that mirrors actual business operations. This process verifies that all front-end interfaces, security gates, cloud configurations, and operational constraints function together smoothly without system failures, resource conflicts, or performance drops.
- **How it Works in Our System:** The researchers performed complete system testing by simulating full operational days at C&N Bakery. The team ran through complete transactional

cycles end-to-end: a researcher acting as a customer logged into a new profile, applied the budget filters (~~P~~0-~~P~~199, ~~P~~200-~~P~~500, ~~P~~501+), assembled a custom pastry order with uploaded attachments, and submitted a payment type. The team then checked the **Administrative Dashboard** to confirm the order card instantly populated the *Pending* tab of the master tracking table, and that the high-level **Dashboard Summary Cards** updated their real-time aggregation counters.

A key part of system testing involved verifying the application's core workflow limit. Researchers simulated high holiday traffic by submitting three separate custom cake orders for the exact same calendar date. The system was monitored to verify that upon the successful database entry of the third order, the **Smart Scheduling Matrix Calendar** immediately locked out that specific date across the cloud network, changing its color state and blocking subsequent customer accounts from selecting it at checkout.

4. User Acceptance Testing (UAT) Phase

- **General Definition:** The final evaluative phase of the testing lifecycle where the live, fully functional software application is handed over to the primary business stakeholders, internal operational staff, and everyday consumers. These users test the platform under real-world conditions to evaluate its overall usability, design clarity, operational efficiency, and practical value before formal launch.
- **How it Works in Our System:** The researchers conducted a formal User Acceptance Testing session inside C&N Bakery, involving the business owner, the kitchen bakers, the counter clerks, and a select group of ten regular customers. The customers were given mobile devices and asked to navigate the storefront catalog, utilize the budget sorting tools, and submit custom order profiles without developer assistance. Simultaneously, the bakery owner and kitchen staff used the

backend dashboard to process these incoming test requests. The staff tested the usability of the **split-screen layout blueprint**, verifying they could easily read the customized cake parameters on one side of their screen while examining the design photo on the other. They also tested the accuracy of the **Payment Verification Toggle** by reviewing uploaded test GCash slips and manually approving payments to watch the **Data Visualization Suite** refresh its vertical and horizontal charts.

Feedback collected during this phase allowed the researchers to adjust button dimensions, modify text contrast, and fine-tune form placeholders, ensuring the completed web platform was comfortable, intuitive, and ready for daily use by the bakery team.

3.7 Evaluation Criteria

To measure the overall success, operational readiness, and software quality of the developed platform, the system was assessed using a standardized set of five technical evaluation criteria. For an engineering project to be considered a viable solution for a real-world business, it cannot merely exist as a collection of features; it must be graded on how accurately it runs, how easily users can navigate it, how securely it handles data, and how quickly it processes business workflows.

The researchers evaluated the application against the following five core pillars to ensure the final product delivers a reliable, production-ready experience for C&N Bakery.

1. Functionality

- **General Definition:** This criterion assesses whether the software application executes every single programmed capability, automated business rule, database calculation, and user feature exactly as specified in the project requirements without logic gaps or processing failures.
- **How it Works in Our System:** The platform's functional capabilities were divided into two main areas and evaluated to ensure complete operational accuracy:
 - **Customer-Facing Operations:** The system was checked to ensure that the online storefront menu correctly applies the range-bound price category filters (**₱0-₱199, ₱200-₱500, and ₱501+**) to isolate products instantly. The **Custom Cake Builder** was evaluated to confirm that its automated pricing scripts compute totals correctly based on user configurations. Furthermore, tests verified that the **Smart Scheduling Matrix Calendar** accurately blocks dates upon hitting the 3-order cap, that the real-time tracking interface displays live order steps, and that the automated

email module sends prompt, itemized notifications whenever an order is confirmed, down payments are approved, or an item is marked for pickup or delivery.

- **Administrative Operations:** The **Administrative Dashboard** was evaluated on its ability to handle backend tasks smoothly. The researchers verified that the instant database text search bar filters through large order tables without missing records, the quick-action *In-Stock* switch toggles pastry availability on the front-end instantly, and the payment verification toggle updates order financials immediately. Finally, the team evaluated the **7-stage production pipeline framework** (*Pending, Confirmed, Baking, Ready, Completed, Cancelled, and the Global View Container*), verifying that order cards move smoothly across these stages inside the interactive **split-screen modal workspace** while updating the underlying **Sales Analytics Suite** summary cards and chart displays.

2. Usability

- **General Definition:** This criterion measures how user-friendly, intuitive, and visually clear the user interface (UI) is. It ensures that users with varying levels of technical experience can easily complete tasks, find information, and navigate the system without confusion or extensive training.
- **How it Works in Our System:** Usability was evaluated from two distinct user perspectives to ensure a low learning curve across the board:
 - **For Customers:** The storefront layout was designed with a simple, clean interface that allows users—even those who are not tech-savvy—to browse items, filter by budget, customize a cake, upload a GCash receipt image, and check out within minutes. Clear layout markers, step-by-step forms, and system alerts guide the buyer naturally through the checkout process.
 - **For Administrators:** Because a busy bakery kitchen is a fast-paced environment, the backend dashboard layout avoids cluttered menus. The researchers focused

heavily on the usability of the interactive **split-screen modal workspace**. By keeping customer text choices on the left and reference photos on the right, bakers can review complicated custom designs at a glance without having to jump between tabs, lowering clerical strain during hectic baking shifts.

3. Reliability

- **General Definition:** This criterion evaluates the long-term stability, consistency, and resilience of the software application under daily use. It ensures the system operates continuously without unhandled system crashes, server drops, or database errors, and that it preserves data integrity even when unexpected network disconnects occur.
- **How it Works in Our System:** The reliability of the platform was evaluated by stress-testing the live cloud connections between the front-end user interfaces and the cloud servers. The application was checked to ensure that multiple customers submitting custom orders simultaneously did not cause database lockups, row overwrites, or broken order numbers.

Because the platform relies on the **Firestore Realtime Database**, reliability evaluations also confirmed that if a user experiences a temporary drop in cellular internet while uploading a heavy cake reference image or a GCash receipt, the system handles the network error gracefully. Rather than crashing the web page or dropping the order text, the application holds the text inputs in a temporary memory cache, pauses the request, and retries the secure cloud upload once the connection stabilizes, preventing data loss for both the customer and the business ledger.

4. Efficiency

- **General Definition:** This criterion measures the speed, processing velocity, and resource usage of the system when handling tasks. It ensures that database queries, algorithm calculations, search actions, and data visualizations display almost instantly, significantly reducing waiting times compared to old paper methods.

- **How it Works in Our System:** The system's performance was measured by timing how long it takes to process core business tasks. The backend was optimized so that when an administrator types a customer name into the instant text search bar, the system filters through database rows and renders the matching results in under a second. Efficiency was also measured within the **Sales Analytics Suite**. When the bakery owner opens the main dashboard, backend aggregation scripts quickly process the entire payment history table. The platform calculates total revenues and transforms thousands of raw data points into visual dashboard summary cards, vertical monthly revenue charts, and horizontal top product graphs almost instantly. By eliminating the need to add up columns by hand in a paper book, the platform drastically cuts down weekly administrative workloads, allowing the owner to make fast, data-driven business decisions.

5. Security

- **General Definition:** This criterion assesses the protective measures built into the software to safeguard private user profiles, sensitive transactional data, and corporate records against unauthorized access, malicious data injections, and digital data leaks.
- **How it Works in Our System:** The researchers evaluated the platform's security layers at multiple entry points. First, user accounts were checked to verify that password fields undergo strong **cryptographic password hashing** before being stored, ensuring that raw passwords are never saved as plain text in the cloud database. Second, all data forms were checked for rigorous input validation; the fields block malicious script strings and invalid characters, keeping the database safe from injection attacks. Finally, because C&N Bakery handles digital down payments, the system's **secure QR code handling** and file attachment system were thoroughly vetted. When a customer uploads a GCash payment receipt screenshot, the file is automatically renamed with a unique, encrypted string and stored within an isolated, private cloud bucket. This structure ensures that financial transaction images are protected from

external access and can only be viewed by authenticated administrators through the secure dashboard layout.

Summary of Chapter III

This chapter outlined the systematic approach used to design, develop, test, and evaluate the **Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics** for C&N Bakery. By grounding the system requirements in triangulated data gathering techniques (interviews, observation, and document analysis), the researchers mapped out the exact operational bottlenecks of the business.

The system was then built using an agile, multi-phase testing lifecycle—running through unit, integration, system, and user acceptance testing—to eliminate software bugs before launch. Finally, by evaluating the platform against strict criteria for functionality, usability, reliability, efficiency, and security, the researchers verified that the completed web application acts as a practical, dependable, and highly secure tool tailored to streamline everyday business for C&N Bakery.

SOLUTION FOR OUR STATEMENT OF THE PROBLEM

1. Customer-Facing Interface and Customization Issues

Research Question 1: *How can a web-based user interface be designed to streamline product navigation, provide clear product descriptions, and completely eliminate the business's reliance on manual social media chat threads for viewing available products?*

SOLUTION :

To fix this issue, our Capstone project builds an online store catalog for the bakery's products, like cakes, cupcakes, and traditional kakanin. Instead of making customers wait for someone to reply on social media, the website automatically shows up-to-date items, descriptions, and prices directly to the user.

Because it runs on a web server, the system can display the menu to hundreds of customers at the exact same time. This means the staff no longer has to stop their work to type out prices over chat. To make browsing even easier, we added simple price filters (₱0-₱199, ₱200-₱500 , ₱501) and a Frequently Asked Questions (FAQ) section. This setup gives customers all the information they need instantly, cuts down on message backlogs during busy holiday rushes, and lets people look through the menu whenever they want without sending a single text message.

Research Question 2: *How can we build a web system that removes manual pricing guesswork by giving customers instant, automated total costs based on the specific cake size, flavor, layers and design options they choose online?*

SOLUTION:

To solve this problem, our Capstone project creates an automated pricing system directly inside the website's database. We assign

fixed price values to every single choice a customer can make. For example, the system knows exactly how much to charge for a specific cake size, a certain flavor, extra cake layers, or an intricate frosting design.

When a customer uses the online custom cake builder, the website acts like an instant calculator. As they click through their options, the system automatically adds up the costs in the background and shows the updated total price on their screen instantly. This removes all mental math and guesswork for the staff. It guarantees that every price quote is 100% accurate, gives the customer complete price transparency before they check out, and ensures the bakery never loses money due to human math errors or employee fatigue.

Research Question 3: How can we design our database and web forms so that the customer's cake design choices, uploaded reference photos, and special text requests are saved accurately and shown directly to the baking staff without errors?

SOLUTION:

To solve this problem, our Capstone project builds a **4-choice custom cake builder** with built-in form controls. Instead of letting customers type out random, unstructured requests in a chat thread, the website provides specific drop-down menus and buttons for their selections (such as cake size, flavor, layers, and design themes). The form also includes an image upload button for reference photos and a strict text box for the cake dedication text.

The moment the customer hits checkout, all of these choices are locked into a permanent, digital data file in our cloud database. This data goes straight to the administrative dashboard without any employee needing to copy or rewrite it. The kitchen staff can then view the exact design choices, text, and uploaded photos clearly on a screen. By removing the need for handwritten slips or chat logs, the system ensures that what

the customer clicks on the website is exactly what the baker sees in the kitchen, completely eliminating misread orders and ingredient waste.

Research Question 4: How can user-generated feedback systems and structured informational modules be effectively integrated into the web application to assist new users in making informed purchasing decisions?

To solve this problem, our Capstone project builds dedicated **customer review modules** and an **interactive Frequently Asked Questions (FAQ) section** directly into the web application. Instead of forcing new customers to blindly trust the business, the website provides a public space where verified buyers can leave star ratings and written feedback about their orders. This data is pulled automatically from the database and displayed directly on the shop's homepage and product catalog.

Additionally, the system includes structured information tabs that answer common questions instantly, covering topics like allergic ingredients, lead times, delivery limits, and step-by-step payment instructions. By creating this official information hub, the system removes customer hesitation, builds instant trust through community proof, and gives new users all the reassurance they need to confidently place an order and send down-payments without needing to ask for basic details over chat.

2. Administrative, Scheduling, and Kitchen Operations Failures

Research Question 5: How can we build an automatic online calendar that tracks and limits daily orders so the bakery never accepts more cakes than the kitchen can handle?

SOLUTION:

To solve this problem, our Capstone project builds a **Smart Scheduling Matrix Calendar** directly into the system's database. Instead of relying on a paper notebook, the system is programmed

with a strict daily capacity rule. For example, management can set the system to only accept a maximum number of custom cakes per day based on their kitchen workload.

The moment a customer tries to book a specific date on the website, the system automatically checks how many orders have already been locked in for that day. If the limit is reached, the calendar instantly blocks that date and gently asks the user to pick a different available slot. Because this digital calendar updates in real time, it acts as a single source of truth for both online customers and walk-in staff, completely preventing double-bookings and protecting the kitchen from stressful work overloads.

Research Question 6: *How can an order monitoring dashboard be structured to replace manual notebooks and provide administrative personnel with real-time visibility over multi-stage production workflows?*

SOLUTION:

To solve this problem, our Capstone project builds a **4-Stage Kitchen Stepper Dashboard** directly into the administrative control desk. Instead of shouting across the room or reading paper cards, the system provides an interactive, visual pipeline screen split into clear milestones: *Processing, Baking, Icing/Styling, and Ready for Pickup.*

As the kitchen staff works on a cake, they can simply tap a button on a tablet or screen to move the order to the next stage. This action instantly updates the entire system. Front counter clerks and managers can see the exact progress of any cake immediately on their own screens without interrupting the bakers. Furthermore, this real-time update syncs with the customer's account view online, letting the buyer know exactly when their custom order is being baked or when it is ready to be collected, saving time for everyone.

Research Question 7: How can we build automatic date checks and warning alerts into the system so that staff members are blocked from accidentally preparing or releasing orders too early?

SOLUTION:

*To solve this problem, our Capstone project builds **Intelligent Date Validation Guards** directly into the administrative workspace and kitchen display. Instead of relying on a human to correctly read an offline paper note, the system treats dates as interactive safety triggers.*

First, the system automatically sorts the kitchen dashboard by priority, only showing orders that match the current day's production window. Second, if an employee tries to prematurely mark a future order as "Ready for Dispatch" or clicks to print a delivery receipt days ahead of schedule, the interface instantly blocks the action. It triggers a clear visual pop-up alert on the screen saying: "Warning: This order is scheduled for a future date." This digital boundary forces the user to double-check the transaction, completely eliminating accidental early preparations and preventing logistical confusion

Research Question 8: How can the system instantly save and update transaction records the moment a delivery or customer pickup is completed?

SOLUTION:

*To solve this problem, our Capstone project builds an **Automated Transaction Archiving Pipeline** that connects the kitchen's progress tracker directly to the sales database ledger.*

The moment the kitchen staff or delivery personnel taps the final stage button ("Order Picked Up / Delivered") on the dashboard, the system triggers two automatic background actions without requiring any manual paperwork. First, it moves the order into a permanent, locked digital archive file. Second, it instantly updates the bakery's electronic sales counter and balances the day's revenue logs in real time. By automating this data synchronization, the platform removes the need for manual

record books or follow-up phone calls, completely eliminates the risk of missing transaction slips, and ensures that daily financial records are perfectly accurate and audit-ready at the close of every business day.

3. Data Privacy, Analytics, and Financial Recording

Research Question 9: How can we build secure login systems and account restrictions within the website to protect private customer accounts and keep the bakery's financial records safe from unauthorized viewing or data loss?

SOLUTION:

To solve this problem, our Capstone project implements a **Role-Based Authentication and Secure Access Gateway** directly into the system design. Instead of leaving data out in the open or sharing everything on one device, the system requires users to log in with secure, password-protected accounts that are strictly split into roles: *Customers, Kitchen Staff, and The Bakery Owner.*

Each role has its own digital boundaries. Customers can only view and manage their own private profile information, order history, and contact details. Kitchen staff can only access the production dashboard to update baking progress. The bakery owner is the only user with the security permissions to view the master financial ledgers, system settings, and complete sales archives. To prevent data disasters, all data is stored on a secure, encrypted cloud database that creates automatic backups, ensuring that even if a physical device drops or breaks in the kitchen, the bakery's records remain 100% safe, private, and fully recoverable.

Research Question 10: How can we build an online payment ledger and receipt upload feature to automatically track customer payments and eliminate manual accounting mistakes?

SOLUTION:

*To solve this problem, our Capstone project builds a **Digital***

Payment Tracker and Receipt-Matching Interface into the system's checkout pipeline. Instead of sending screenshots to a separate chat app, customers upload their GCash or PayMaya transaction receipts directly onto the website while placing their order.

The moment a receipt is uploaded, the system automatically binds that image file to the customer's specific order ID in the database. On the management side, the bakery owner opens a single dashboard where the customer's design choices, balance details, and uploaded payment slip are displayed side-by-side on one screen. The owner can click "Verify" to update the transaction instantly. The system then automatically subtracts the down-payment from the total cost and locks in the remaining balance. This digital linkage keeps the books perfectly balanced, eliminates loose screenshots on mobile phones, and prevents uncomfortable payment arguments at the pickup counter.

Research Question 11: How can we turn saved sales data into visual, real-time charts on a dashboard to help the bakery owner see profit margins and make smarter business decisions?

SOLUTION:

To solve this problem, our Capstone project builds an automated **Business Intelligence and Sales Analytics Dashboard** directly into the admin panel. Instead of making the owner manually sort through hundreds of paper slips at the end of the month, the website's database processes every completed sale instantly.

The system takes this raw information and automatically transforms it into simple, easy-to-read visual charts and graphs that update in real time. When the owner opens the dashboard, they can instantly see exact business metrics, including:

- **Product Performance:** Which items (cakes, cupcakes, or kakanin) are selling the fastest.
- **Peak Ordering Trends:** What times of day or days of the week get the most traffic.

- **Pricing Tier Breakdown:** Which specific budget bracket (P0-P199 , P200-P500 , or P501+) brings in the highest total revenue.
- **Order Health Ratios:** A clear look at completed orders versus cancelled ones.

By providing these instant visual charts, the system completely removes manual math guesswork, protects the owner from data blindness, and gives management the clear insights needed to stock the right ingredients, prepare for busy holiday rushes, and maximize the bakery's profits.